

Beyond Nationality

Firm Foreignness, Employment, and Industrial Subsidies

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Abstract

Governments routinely subsidize foreign firms, even though industrial policy is justified as advancing national economic interests. How do governments choose among globally organized firms, and do citizens evaluate potential recipients using the same logic? I argue that firm nationality and reliance on foreign suppliers represent distinct dimensions of foreignness, while employment provides a visible domestic economic contribution that can make support for a foreign firm easier to justify. I examine this argument by combining evidence on actual government allocation with two preregistered survey experiments on nationally representative samples totaling 4,000 U.S. adults. In firm-level data on U.S. subsidies from 2000 to 2023, foreign firms receive fewer subsidies and lower total amounts, but this disadvantage narrows among more labor-intensive firms. Citizens also distinguish ownership, workforce location, and supplier origin, but apply a different weighting. Foreign workforces and suppliers substantially reduce public support, while the ownership effect is smaller and less consistent. Together, the studies show that government allocation and public evaluation share attention to the same firm characteristics but do not follow the same decision rule. By connecting actual subsidy allocation to its potential public foundations, the paper shows how industrial policy navigates the tension between national objectives and globally organized firms.

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Introduction

In 2022, the state of Georgia offered \$1.5 billion in incentives to Rivian, an American electric-vehicle manufacturer, for a new \$5 billion plant expected to create 7,500 jobs. Just months later, the state offered an even larger package, worth up to \$1.8 billion, to Hyundai, a Korean automaker, for an electric-vehicle and battery complex expected to create more than 8,000 jobs (Associated Press, 2022; Kanell, Trubey, and Hansen, 2022). Despite Hyundai’s foreign nationality, Georgia officials emphasized the jobs, investment, and broader manufacturing activity the project would bring to the state (Office of Governor Brian P. Kemp, 2022). The two cases illustrate a broader puzzle in the politics of industrial policy: if governments favor domestic firms, why do they sometimes provide comparable, or even greater, support to foreign firms?

Industrial policy has returned to the center of economic policymaking, as governments increasingly subsidize firms to promote strategic industries, technological competitiveness, and economic security (Aiginger and Rodrik, 2020; Juhász, Lane, and Rodrik, 2024; Breznitz and Gingrich, 2025). Yet globalization has blurred the distinction between “domestic” and “foreign” firms. A foreign manufacturer may employ thousands of domestic workers and produce locally, while a domestically headquartered firm may rely heavily on overseas production and global supply chains (Helpman, Melitz, and Yeaple, 2004; Bernard et al., 2007; Gereffi, 2014; Kim and Osgood, 2019; Osgood, 2018). As nationality becomes increasingly decoupled from the location of economic activity, it is less clear whether governments should prioritize where a firm is headquartered or the economic benefits it generates at home.

Existing research offers only a partial answer. Scholarship on industrial policy and economic-development incentives explains why governments target particular industries, locations, or politically valuable firms (Bartik, 2019; Jensen and Malesky, 2018; Jensen, Findley, and Nielson, 2020; Slattey, 2025, 2026), while research on multinational enterprises and economic nationalism examines the political consequences of foreign nationality (Pandya, 2010; Tingley et al., 2015; Mansfield and Mutz, 2009; Mutz, Mansfield, and Kim, 2021; Kim and Siegel, 2024). Yet we know much less about how governments and the public evaluate firms when nationality and domestic economic contribution point in different directions.

I argue that governments distinguish between firm foreignness and domestic economic contribution. Firm foreignness has two distinct dimensions. Firm nationality indicates where the company is headquartered and where corporate control resides, while foreign supplier reliance indicates how much of its upstream production is located abroad. Domestic employment plays a different role. It provides a visible and locally attributable benefit

that can make support for a foreign firm easier to justify. Employment therefore does not constitute another dimension of foreignness. Instead, it can moderate the political disadvantage associated with foreign nationality.

This distinction generates two sets of expectations. First, foreign nationality and reliance on foreign suppliers may each reduce government support because both create concerns that the benefits of subsidies will flow abroad. Second, the nationality penalty should not be uniform. Governments should be more willing to support foreign firms when their operations generate visible employment. Globalization also raises a related question that existing research has largely overlooked: whether citizens evaluate the same firm characteristics in the same way. Governments routinely justify subsidies to foreign firms by emphasizing domestic jobs. If citizens share this compensatory logic, an American workforce should reduce the ownership penalty. Alternatively, citizens may treat ownership, workforce location, and supplier origin as separate considerations.

To examine both sides of this political process, I combine two complementary studies. Study 1 examines how governments actually allocate industrial subsidies across firms that differ in nationality, employment orientation, and foreign supplier reliance. Study 2 uses survey experiments to identify how citizens respond to the same distinctions when deciding whether a firm deserves public support. The observational analysis establishes patterns in government behavior but cannot directly identify the public logic behind them. The experiments isolate citizens' responses to firm characteristics but do not show how governments allocate subsidies in practice. Together, the studies allow me to assess whether actual allocation and public evaluation reflect a common political logic.

The analyses produce three main findings. First, foreign firms receive fewer subsidies and lower total subsidy amounts than comparable U.S. firms, but this nationality penalty is significantly attenuated among more labor-intensive firms. Second, foreign supplier reliance does not reduce government support, indicating that governments distinguish foreign nationality from foreign supply-chain exposure. Third, citizens respond strongly to workforce and supplier location, but the ownership effect is smaller and less consistent. An American workforce does not systematically reduce the ownership penalty, and foreign ownership and foreign sourcing do not significantly reinforce one another. Government allocation is therefore consistent with employment orientation partially compensating for foreign nationality, whereas citizens largely evaluate ownership, workforce location, and supplier origin independently.

These findings contribute to research on industrial policy, globalization, and economic

nationalism in three ways. First, the paper shifts attention from sectoral targeting to how governments distinguish among firms within the same industry. Second, it separates two dimensions of firm foreignness, nationality and foreign supplier reliance, from employment as a domestic economic contribution, and shows that these characteristics have different political consequences. Third, and most importantly, the paper connects government allocation to the public legitimacy of industrial policy. Existing research generally examines either how governments distribute economic benefits or how citizens evaluate international economic policies. By examining both sides using comparable firm characteristics, this paper shows that governments and citizens distinguish nationality, employment, and supplier relationships but weigh and combine them differently. Government allocation is consistent with employment orientation mitigating the nationality penalty. Citizens strongly favor an American workforce, but do not use it to revise their response to foreign ownership. The political logic reflected in government allocation is therefore only partially shared by the public.

Theory

Industrial policy in a globalized economy

Industrial policy has re-emerged as a central instrument of economic policy. Governments increasingly intervene to promote strategic industries, accelerate technological innovation, strengthen supply-chain resilience, and enhance national competitiveness through targeted support for particular firms, sectors, and technologies (Lane, 2020; Naughton, 2021; Juhász, Lane, and Rodrik, 2024). Although industrial policy encompasses a wide range of instruments, including tariffs, procurement, and sector-wide incentives, this paper focuses on firm-level production subsidies. Unlike broader policy tools, firm-specific subsidies require governments to identify individual firms that merit public support. Every subsidy decision therefore reflects an implicit judgment about which firms are expected to advance national economic objectives.

These judgments have become increasingly difficult under globalization. Over the past several decades, multinational production and global value chains have fundamentally reshaped the organization of firms (Helpman, Melitz, and Yeaple, 2004; Gereffi, 2014; Kim and Osgood, 2019; Osgood, 2018). Nationality, production, employment, and sourcing are no longer concentrated within a single country, but are instead distributed across multiple jurisdictions. As a result, the traditional distinction between domestic and foreign firms has

become increasingly blurred. A foreign firm may employ thousands of domestic workers, purchase extensively from domestic suppliers, and produce almost entirely within the host economy. Conversely, a domestically headquartered firm may conduct much of its production abroad while relying heavily on imported inputs. Nationality alone therefore provides an increasingly incomplete picture of where a firm's economic activities, and the benefits they generate, are actually located.

This transformation poses a fundamental challenge for industrial policy. Governments continue to justify subsidies in terms of national economic development, emphasizing domestic job creation, investment, and industrial competitiveness. Yet these objectives no longer map neatly onto firm nationality. A foreign firm may create substantial domestic employment, while a U.S. firm may direct much of its purchasing to suppliers abroad. Policymakers must therefore distinguish between how a firm is connected to foreign economies and what economic benefits it produces at home.

I separate these considerations into two parts. First, *firm foreignness* has two dimensions in this paper. Firm nationality identifies where the firm is headquartered and where corporate control resides. Foreign supplier reliance identifies the extent to which the firm's upstream production relationships are located abroad. The two dimensions need not coincide: a U.S. firm can depend heavily on foreign suppliers, while a foreign firm can source extensively within the United States. Second, *domestic economic contribution* refers to benefits that the focal firm produces within the United States. I focus on employment because jobs are visible, geographically concentrated, and readily attributable to the firm receiving the subsidy.

These concepts play different theoretical roles. Foreign nationality and foreign supplier reliance provide separate reasons to worry that the benefits of public support may accrue abroad. Employment provides a countervailing domestic benefit. It does not make a firm less foreign. Instead, it can make a foreign firm more attractive to policymakers by giving them a concrete domestic outcome with which to justify public support. The argument therefore does not place nationality, suppliers, and employment on a single continuum. It asks how governments weigh two dimensions of foreignness and whether employment moderates the penalty associated with firm nationality.

Foreign nationality as a baseline political liability

Although globalization has weakened the correspondence between corporate nationality and domestic economic activity, nationality remains one of the most salient characteristics of firms in industrial policy. A firm's national affiliation provides a visible cue about where

corporate control resides and where the returns to public investment may ultimately accrue. Governments allocating public resources therefore cannot ignore nationality, even when it provides only a partial picture of a firm’s contribution to the domestic economy.

Foreign nationality creates political costs for policymakers through several mechanisms. Economically, subsidizing a foreign firm raises the possibility that part of the returns generated by public investment will accrue abroad rather than remaining within the domestic economy. Politically, national affiliation is highly visible and easily understood by voters, making it a focal point for criticism when governments support foreign firms. Opponents can readily portray such subsidies as transfers of taxpayer resources to foreign companies, regardless of where production occurs or who ultimately benefits from the investment.

This political sensitivity is longstanding. When Honda opened its automobile assembly plant in Marysville, Ohio in the early 1980s, critics warned that Japanese investment risked turning parts of American manufacturing into “manufacturing colonies of Japan,” despite the jobs and production the plant generated locally (Behr, 1987). The criticism focused on Honda’s Japanese national affiliation rather than where production took place or who benefited from it locally.

These concerns are further reinforced by the political relationships surrounding industrial policy. Domestic firms generally have stronger local political and economic networks, making them better positioned to lobby governments, cultivate relationships with policymakers, and provide visible political or economic returns following public support (Siegel, 2007; Johns and Wellhausen, 2021; Kim and Siegel, 2024). Foreign firms, in contrast, face greater legal, institutional, and informational barriers to developing these relationships.

For these reasons, foreign nationality is expected to create a baseline political disadvantage in the allocation of industrial subsidies. Importantly, this expectation does not imply that governments categorically avoid supporting foreign firms. Contemporary industrial policy routinely relies on multinational enterprises to achieve strategic objectives, particularly in sectors such as semiconductors, automobiles, and clean energy. The relevant question is therefore not whether governments subsidize foreign firms, but under what conditions they are willing to do so. I argue that nationality establishes the initial political liability against which other characteristics of firms are evaluated. Whether this liability can be overcome depends on the extent to which a firm’s activities generate visible domestic economic benefits, a question developed in the following subsection.

Hypothesis 1. Foreign firms are less likely to receive industrial subsidies than otherwise comparable U.S. firms.

Foreign supplier dependence as a second dimension of foreignness

Firm nationality is not the only way globalization connects firms to foreign economies. Firms also differ in the extent to which they rely on domestic or foreign suppliers. Purchasing inputs from domestic suppliers directs the indirect gains from expanded production toward upstream domestic producers, while foreign sourcing shifts more of these gains abroad. Research on global value chains shows that supplier relationships shape both the distribution of firms' economic activity and the political interests generated by international production (Johns and Wellhausen, 2016; Osgood, 2018; Zeng, Sebold, and Lu, 2020). Foreign supplier reliance is therefore a second dimension of firm foreignness, separate from the nationality of the focal firm.

The political implications of supplier relationships, however, differ from those of firm nationality. Nationality is a visible cue about corporate control and national affiliation. Supplier relationships are harder to observe, and their consequences are dispersed across numerous firms and locations. Politicians can easily identify the nationality of a subsidy recipient, but it is more difficult to explain how support for that firm will affect a complex upstream production network. Foreign supplier reliance should therefore create a weaker and less consistent political penalty than foreign nationality.

Recent debates surrounding Chinese electric vehicle manufacturers illustrate this difference. Political opposition to BYD's prospective expansion into the United States has emphasized the firm's Chinese nationality and associated national-security concerns (Office of Senator Tammy Baldwin, 2024; Reuters, 2026). Although policymakers have also raised concerns about dependence on Chinese batteries and parts, my interpretation of this debate is that nationality and national security have been more prominent than the geographic composition of the firm's supplier network. The contrast illustrates the broader theoretical argument: supplier relationships contribute to how governments evaluate firms, but they are less visible and therefore less politically consequential than nationality itself.

Hypothesis 2. Firms that rely more heavily on foreign suppliers are less likely to receive industrial subsidies than otherwise comparable firms.

Employment as a domestic economic contribution

The preceding arguments identify two dimensions of firm foreignness, but they leave unresolved the central puzzle of contemporary industrial policy: if foreign nationality creates a political disadvantage, why do governments nevertheless subsidize many foreign firms? In

practice, governments frequently support foreign firms when they are expected to generate substantial domestic economic benefits. The relevant question is therefore not simply whether foreign firms are disadvantaged, but what allows some of them to overcome the nationality penalty.

I argue that domestic employment provides the clearest answer. Among the ways firms can contribute to the domestic economy, employment is uniquely visible, geographically concentrated, and politically attributable. New jobs are readily observed by citizens, frequently reported by the media, and routinely highlighted in government announcements of industrial subsidies. Unlike productivity improvements, technological spillovers, or future tax revenues, employment provides policymakers with a concrete and easily communicated justification for public investment. Politicians can credibly claim that subsidy programs create jobs for local workers and strengthen local communities.

This logic is especially important for foreign firms. Because foreign nationality raises concerns that the benefits of public support may ultimately flow abroad, governments require stronger evidence that subsidizing these firms serves domestic interests. Visible employment provides precisely such evidence. By creating jobs within the host economy, foreign firms can demonstrate that at least part of the returns from public investment remain with domestic workers and communities. Domestic employment can therefore provide a political justification that partially offsets the baseline disadvantage associated with foreign nationality.

BMW's manufacturing complex in Spartanburg, South Carolina illustrates this logic. Although BMW is a German firm, the facility has become one of the state's largest employers and is routinely cited by both BMW and state officials for the tens of thousands of jobs it supports directly and indirectly (BMW Group, 2024). Public discussion surrounding government support for the facility has consequently focused far more on its employment contribution than on its foreign nationality. The example illustrates the broader argument developed here: domestic employment does not eliminate the political relevance of nationality, but it can substantially reduce its political costs by providing visible evidence that subsidizing a foreign firm advances domestic economic objectives.

Directly comparing firms by the number of workers they employ is difficult because employment naturally scales with firm size. I therefore operationalize firms' employment orientation using labor intensity, the extent to which production relies on labor relative to capital. Holding firm size constant, labor intensity distinguishes firms for which employment constitutes a larger or smaller component of economic activity. Because the available measure does not identify where employees are located, it should be interpreted as a proxy for

employment orientation rather than a direct measure of domestic employment.

Hypothesis 3. The disadvantage faced by foreign firms is attenuated among more labor-intensive firms.

Labor intensity may also operate through supply chains. Firms connected to labor-intensive foreign suppliers could, in principle, shift employment benefits abroad, raising concerns about whether subsidies support domestic job creation. As with supply-chain nationality more broadly, however, supplier-level employment is less visible and less readily attributable to a subsidy decision than employment at the focal firm.

Hypothesis 4. The negative relationship between reliance on foreign suppliers and subsidy allocation is stronger when those suppliers are more labor-intensive.

Together, Hypotheses 1–4 distinguish two sources of foreign exposure from one countervailing domestic benefit. Firm nationality and foreign supplier reliance capture different dimensions of foreignness. Employment at the focal firm captures a visible economic contribution that can reduce the nationality penalty. Whether citizens draw the same distinctions remains an open question.

Public support for industrial subsidies

Industrial subsidies are financed with public resources and ultimately derive their political legitimacy from public support. Yet ordinary citizens evaluate firms under very different informational conditions from policymakers. Unlike governments, citizens do not negotiate directly with firms, observe firms' investment plans, or evaluate detailed projections of economic impact. Instead, they infer whether a firm deserves public support from a limited set of observable characteristics. The survey experiments make the three relevant characteristics explicit: firm ownership, workforce location, and supplier origin.

Material benefits One possibility is that citizens evaluate firms according to where the expected benefits of a subsidy will accrue. A predominantly American workforce provides a direct domestic economic contribution by creating jobs for American workers. Supplier origin provides different information: it indicates whether the indirect gains from production will flow toward American or foreign upstream firms. Citizens who focus on material benefits should therefore favor firms with American workers and penalize firms that rely primarily on foreign suppliers (Hainmueller and Hiscox, 2010; Mansfield and Mutz, 2009; Owen and

Johnston, 2017; Autor, Dorn, and Hanson, 2013; Kim and Margalit, 2016=7; Pandya, 2010; Lee and Liou, 2022).

Ownership as a symbolic cue At the same time, citizens may attach independent political significance to firm ownership. Industrial subsidies are financed with taxpayer resources and justified as advancing national interests, making ownership a visible cue about whether the recipient belongs to the national economic community. Research on economic nationalism and foreign investment suggests that symbolic concerns about national identity, sovereignty, and economic control frequently shape attitudes toward international economic policies alongside material considerations (Mansfield and Mutz, 2009; Mutz, Mansfield, and Kim, 2021; Powers, Reifler, and Scotto, 2021; Feng, Kerner, and Sumner, 2021; Bankert, Powers, and Sheagley, 2023; Chan and Meunier, 2022; Tingley et al., 2015; Carnegie and Gaikwad, 2022). From this perspective, an American workforce can increase a firm’s attractiveness by generating a domestic economic contribution, but it does not change the fact that a foreign-owned firm remains foreign. Foreign supplier reliance provides an additional indication that economic activity extends beyond the United States.

These arguments generate expectations about both the independent and combined effects of the three characteristics. Citizens should be less supportive of foreign-owned firms, firms with foreign workforces, and firms that rely on foreign suppliers. The interactions test two different theoretical relationships. In Experiment 1, an American workforce may compensate for foreign ownership by giving the firm a visible domestic economic contribution. In Experiment 2, foreign ownership and foreign sourcing may reinforce one another because they represent two distinct dimensions of firm foreignness.

Survey Hypothesis 1s. The public will be less supportive of providing subsidies to foreign-owned firms than to American-owned firms.

Survey Hypothesis 2sa. The public will be less supportive of providing subsidies to firms that employ primarily foreign workers than to firms that employ primarily American workers.

Survey Hypothesis 2sb. The public will be less supportive of providing subsidies to firms that source primarily from foreign suppliers than to firms that source primarily from American suppliers.

Survey Hypothesis 3s. The negative effect of foreign ownership on public support will be smaller when a firm employs primarily American workers than when it employs primarily foreign workers.

Survey Hypothesis 4s. The negative effect of foreign ownership on public support will be larger when a firm relies primarily on foreign suppliers than when it relies primarily on American suppliers.

Hypothesis 3s is the direct public-opinion counterpart to the employment argument developed for government allocation. Hypothesis 4s asks whether the two dimensions of foreignness have cumulative effects. The interactions are mathematically similar, but their theoretical meanings differ. An American workforce supplies a domestic economic contribution that may offset the ownership penalty; American suppliers indicate the absence of a second form of foreign exposure.

The following sections evaluate these expectations using two complementary empirical strategies. The first examines how governments allocate industrial subsidies across firms differing in nationality, foreign supplier reliance, and employment orientation. The second tests whether citizens distinguish ownership and supplier-based foreignness from the domestic employment contribution in two preregistered survey experiments.

Study 1: Government Allocation of Industrial Subsidies

Data and sample

To evaluate Hypotheses 1–4, I examine the allocation of industrial subsidies to firms in the United States from 2000 to 2023. The United States provides a particularly useful setting because it combines extensive multinational firm activity and cross-border production (Kose et al., 2017) with active subsidy competition and economic-development programs at the federal, state, and local levels (Jensen and Malesky, 2018; Jensen, Findley, and Nielson, 2020; Slaterry, 2025). This creates substantial variation not only in whether firms are headquartered in the United States or abroad, but also in the extent to which their production, employment, and supplier relationships are connected to the U.S. economy.

The analysis combines three principal data sources: Subsidy Tracker, Compustat, and Orbis. Subsidy Tracker provides information on firm-specific government assistance, including grants, tax credits and abatements, subsidized loans, and other forms of direct economic-development support. Compustat supplies firm financial characteristics, employment, industry, and headquarters information. I also use Compustat Customer Segments to recover major customer relationships and invert these reported seller–customer links to identify upstream suppliers. Orbis provides additional information on firms’ nationality and

corporate structure. Because these sources use different firm identifiers, I link records using GVKEY, ISIN, ticker symbols, and firm names, supplemented by manual verification through WRDS and Capital IQ.¹

The resulting dataset is organized at the firm-year level and covers publicly traded firms observed between 2000 and 2023. This structure makes it possible to compare firms operating in the same year, state, and industry while linking government support to characteristics that are rarely observed together in existing studies of industrial policy: firm nationality, employment intensity, and the geographic composition of major supplier relationships.

The dependent variables capture two dimensions of industrial support. The first is the number of subsidy awards received by firm i in year t . The second is the total dollar value of those awards. Considering both outcomes is important because governments may differentiate among firms at different stages of subsidy allocation. Foreign firms could be excluded from subsidy programs altogether, for example, or they could gain access to support but receive fewer or smaller awards once selected. The analyses therefore examine both the frequency and the monetary scale of industrial support.²

Figure 1 shows the geographic distribution of firms and subsidy receipt across the United States. Subsidized firms are geographically dispersed rather than concentrated in a small number of states, reflecting the combination of federal, state, and local programs captured in the data.

Measures

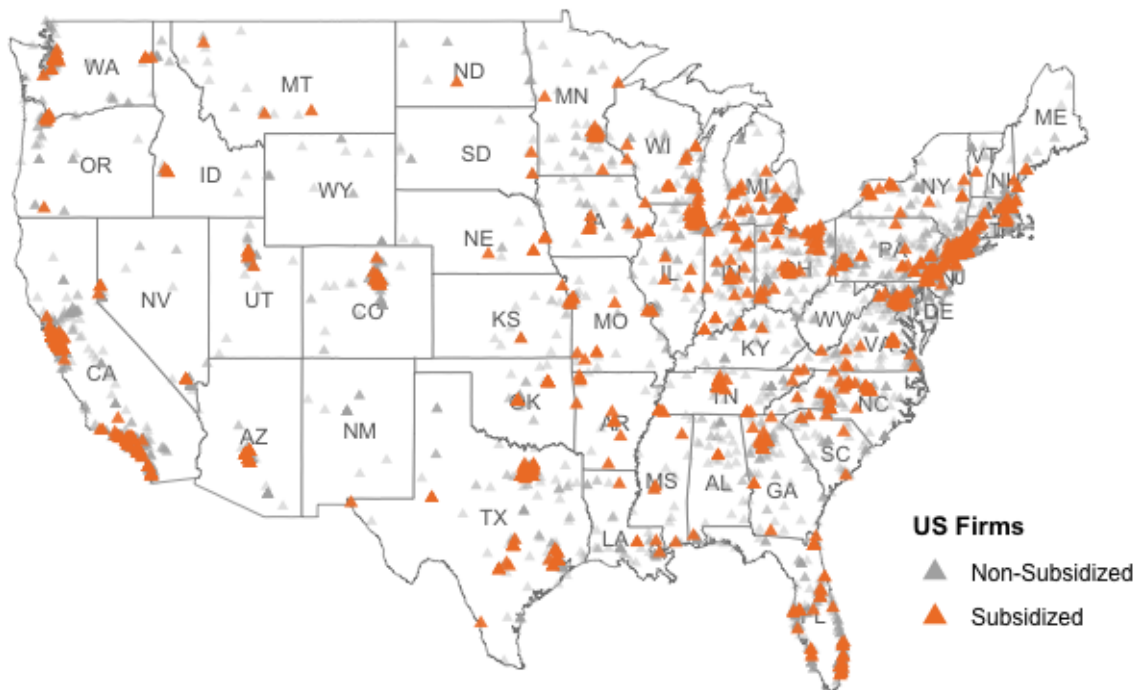
Firm nationality The first dimension of firm foreignness is nationality. I construct a binary measure, Foreign firm, coded one when a firm is headquartered outside the United States and zero when it is U.S.-headquartered, using Compustat headquarters and fiscal-country information. This measure captures the nationality distinction at the center of Hypothesis 1.

Foreign firms constitute a substantial share of the sample. As Table 1 shows, 23.0% of firm-year observations are foreign, but foreign firms account for only 15.4% of subsidy recipients. The unconditional receipt rate also differs sharply by nationality: 10.1% of U.S.-firm years receive at least one subsidy, compared with 6.2% of foreign-firm years. Among recipients, U.S. firms also receive somewhat more awards and larger average subsidy amounts.

¹ The matching procedure combines automated identifier-based matching with manual verification for records that could not be linked uniquely.

² State governments account for 73.3% of subsidy projects in the data, while federal programs account for 54.3% of total subsidy dollars. The theoretical argument concerns how firm foreignness shapes the political allocation of public support and is not limited to a single level of government.

Figure 1: Geographic distribution of firms and industrial subsidy receipt



Note: Points represent firm-year observations located at firms' headquarters. Orange points indicate firms receiving at least one industrial subsidy during the observation year; gray points indicate firms receiving no subsidy.

These descriptive differences do not establish a nationality penalty; foreign and U.S. firms differ in size, industry, geography, and other characteristics, but they provide an initial indication of the pattern examined below.³

Firm labor intensity Hypothesis 3 concerns whether employment orientation attenuates the disadvantage associated with foreign nationality. Raw employment counts are not well suited to this comparison because larger firms mechanically employ more workers. I therefore measure the employment intensity of production as the ratio of employees to annual revenue. Higher values identify firms whose economic activity relies more heavily on labor relative to revenue, while lower values characterize more capital-intensive production. As Panel (a) of Figure 2 shows, labor intensity varies considerably across firms and is strongly right-skewed. I therefore use its logged transformation in the regression analyses.

³ Foreign subsidy recipients are headquartered in 45 countries. Japanese firms receive the largest number of awards, followed by firms headquartered in Germany, Switzerland, and Canada.

Table 1: Subsidy receipt by firm nationality

	All firm-years	U.S. firms	Foreign firms
N	151,338	116,456	34,882
Share of sample	100%	77.0%	23.0%
Share of all recipients	—	84.6%	15.4%
% receiving any subsidy	9.2%	10.1%	6.2%
Avg. # awards among recipients	4.14	4.22	3.69
Mean subsidy amount (recipients only)	—	\$12,861,464	\$11,119,965

This measure should be interpreted as an indicator of the prominence of employment in a firm’s productive activity rather than as a direct count of U.S. workers. Compustat does not identify the location or nationality of every employee within a multinational firm, so the measure cannot establish precisely how many jobs attributable to a particular firm are located in the United States. Its value is instead comparative. Holding firm size, industry, location, and year constant, labor intensity distinguishes firms for which employment constitutes a larger or smaller component of economic activity. The analysis therefore tests whether an employment-oriented production structure moderates the nationality penalty, not whether a measured number of U.S. jobs causes governments to provide subsidies.

The distinction is particularly important for foreign firms. The theoretical argument does not simply predict that labor-intensive firms receive more subsidies on average. Rather, employment orientation should matter especially when foreign nationality creates an initial political liability. The key test of Hypothesis 3 is therefore whether the nationality gap in subsidy allocation narrows as labor intensity increases.

Foreign supplier dependence The second dimension of firm foreignness concerns supply chains. Compustat Customer Segments reports firms’ major customers rather than their procurement transactions. I therefore invert the reported seller–customer links: a firm is identified as an upstream supplier when it reports the focal firm as a major customer. Using the geographic location of these identified suppliers, I construct Foreign supplier share as the share of supplier-linked sales associated with foreign upstream suppliers, weighted by the sales reported for each customer relationship.⁴ The measure ranges from zero, indicating an entirely domestic observed supplier base, to one, indicating that all identified major suppliers

⁴ Compustat customer relationships are derived from firms’ SEC disclosures and linked across firms using verified name-matching procedures. The measure captures disclosed major-customer relationships viewed from the buyer’s side; it is not a comprehensive measure of the focal firm’s procurement (Patatoukas, 2012; Cohen and Li, 2020).

are foreign.

Foreign supplier exposure has a markedly different distribution from firm nationality. Approximately 92% of firm-year observations report no foreign major suppliers. Among the remaining observations, however, reliance on foreign suppliers varies substantially, including firms with partially foreign and entirely foreign observed supplier networks. Panel (b) of Figure 2 displays this variation among observations with positive foreign supplier exposure; the large mass at zero is omitted from the figure for visual clarity.

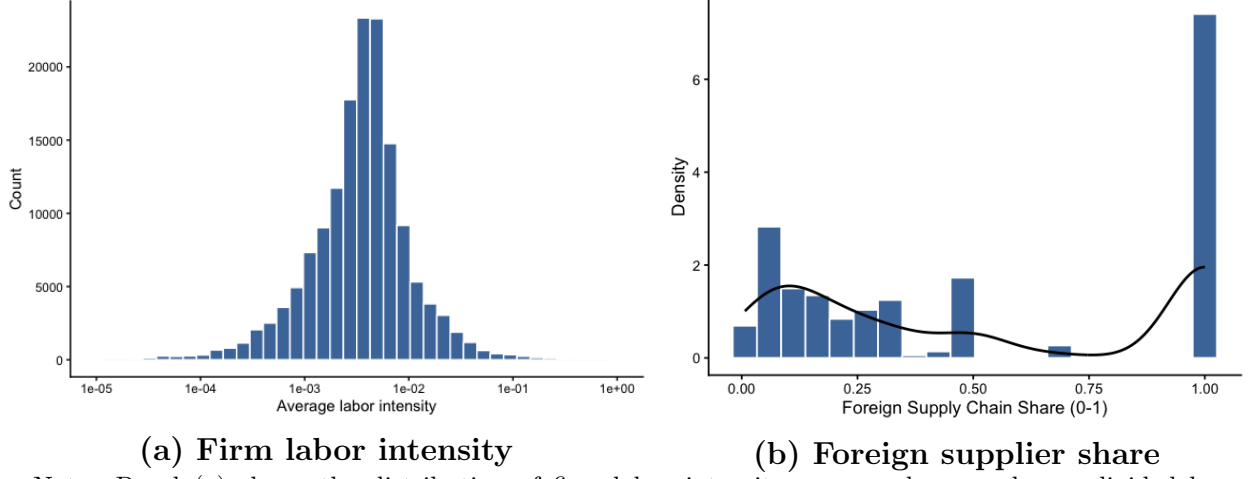
Supplier information is also available for a substantially smaller set of firm-years than nationality and labor-intensity data because disclosure of major supplier relationships is not universal. Models incorporating Foreign supplier share therefore use a more restricted sample than the nationality-only specifications. This difference is important when comparing coefficient magnitudes across models: changes in the foreign-firm coefficient after introducing supplier characteristics may partly reflect changes in sample composition rather than the statistical effect of controlling for foreign sourcing itself.

Supplier labor intensity I also measure the labor intensity of firms' major suppliers using the same employee-to-revenue ratio applied to focal firms. This measure captures whether a firm's supplier network is concentrated among relatively labor-intensive or capital-intensive producers. I interact supplier labor intensity with Foreign supplier share to examine whether the consequences of foreign sourcing depend on the employment intensity of the supplier network.

If governments are sensitive not only to where firms source but also to where employment benefits generated through those supply chains accrue, foreign supplier dependence should be more politically costly when the relevant suppliers are labor intensive. In that case, greater reliance on foreign suppliers implies that a larger share of employment associated with upstream production is located abroad. Because supplier-level employment is less visible and less readily attributable to a subsidy decision than employment at the subsidized firm itself, however, I expect this conditional relationship to be weaker than the interaction between foreign nationality and focal-firm labor intensity.

Controls All specifications account for firm size using information on revenue and employment and include year, state, and four-digit NAICS industry fixed effects. These fixed effects compare firms facing similar temporal, geographic, and sectoral environments, absorbing common shocks to subsidy availability as well as persistent differences across states and industries in the use of industrial policy. Foreign supplier exposure is lagged by one year where appropriate to reduce simultaneity between contemporaneous subsidy receipt

Figure 2: Distribution of firm labor intensity and foreign supplier share



Note: Panel (a) shows the distribution of firm labor intensity, measured as employees divided by annual revenue; the horizontal axis is displayed on a log scale because the measure is highly right-skewed. Panel (b) shows foreign supplier share among firm-year observations with positive foreign supplier exposure. Approximately 92% of firm-year observations report no foreign major suppliers and are omitted from Panel (b) for visual clarity.

and supplier composition.

Empirical strategy

The baseline models estimate the association between firm foreignness and industrial subsidy allocation while holding constant firm characteristics and fixed effects. Let Y_{it} denote either the number or total amount of industrial subsidies received by firm i in year t . The baseline specification is

$$\ln(1 + Y_{it}) = \beta_0 + \beta_1 \text{Foreign firm}_{it} + \beta_2 \text{Foreign supplier share}_{i,t-1} + \gamma \mathbf{X}_{it} + \alpha_t^{\text{Year}} + \alpha_s^{\text{State}} + \alpha_j^{\text{Industry}} + \varepsilon_{it},$$

where $\mathbf{X}_{it}$ contains firm-level controls. The coefficient on Foreign Firm evaluates Hypothesis 1, while the coefficient on Foreign Supplier Share evaluates Hypothesis 2. Because supplier information is available for a more restricted set of firms, I first estimate the nationality relationship on the broader sample and then introduce supplier characteristics.

To evaluate Hypothesis 3, I interact foreign-firm status with logged labor intensity. The quantity of interest is not the coefficient on either term in isolation, but how the predicted nationality gap changes across the observed distribution of labor intensity. I therefore complement the interaction estimates with predicted outcomes for U.S. and foreign firms across

substantively relevant values of labor intensity. To provide substantive estimates on the original outcome scale, I also estimate Poisson pseudo-maximum-likelihood models using the untransformed subsidy outcomes and the same controls and fixed effects.

All baseline models are estimated using OLS with log-transformed outcomes and robust standard errors. The appendix evaluates whether the results depend on this functional form or on treatment of the large number of firm-years receiving no subsidies. These analyses include binary receipt models, zero-inflated negative binomial specifications for subsidy counts, alternative fixed-effects specifications, firm-clustered standard errors, and matching estimators that compare observationally similar foreign and domestic firms.

Results

Firm nationality and industrial subsidy allocation

Table 2 first examines the relationship between firm foreignness and industrial subsidy allocation. Consistent with Hypothesis 1, foreign nationality is associated with substantially less government support. In the broad sample, foreign firms receive significantly fewer subsidies than comparable U.S. firms ($\beta = -0.115$, $p < 0.01$). On the log scale, this estimate corresponds to approximately 11% fewer subsidy awards, conditional on the included controls and fixed effects.

The nationality gap is considerably larger for subsidy amounts. Foreign firms receive lower total subsidy amounts per firm-year than otherwise comparable U.S. firms ($\beta = -0.995$, $p < 0.01$), corresponding to a reduction of roughly 63%. The contrast between counts and amounts is substantively informative. Foreign firms are not categorically excluded from industrial subsidy programs; rather, the nationality disadvantage appears especially pronounced in the overall scale of public support they receive. Because the amount outcome includes firm-years with no subsidies, this difference reflects both access to subsidies and the value of the awards received; it should not be interpreted solely as evidence that foreign firms receive smaller awards conditional on receiving support.

The foreign-firm coefficient becomes considerably larger in the joint specifications that include supplier characteristics. These models, however, are estimated on the substantially smaller set of firms for which major supplier relationships can be observed. I therefore do not interpret the increase in coefficient magnitude as evidence that controlling for foreign sourcing itself reveals a larger nationality penalty. The more conservative conclusion is that the negative nationality relationship persists both in the broader sample and among firms

Table 2: Firm foreignness and industrial subsidy allocation

	1	2	3
	Log Count of Industrial Subsidies		
Foreign firm	-0.115** (0.026)		-0.785*** (0.072)
Foreign supplier share _{t-1}		0.097* (0.034)	0.097** (0.034)
	Log Amount of Industrial Subsidies		
Foreign firm	-0.995** (0.310)		-6.015*** (1.332)
Foreign supplier share _{t-1}		0.733* (0.307)	0.739* (0.307)
Firm size	Yes	Yes	Yes
State FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes

Note: Ordinary least squares estimates with robust standard errors in parentheses.

All models include firm size, year, state, and 4-digit NAICS industry fixed effects.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

with observable supplier networks.

This pattern is robust to several alternative specifications. Binary models of whether a firm receives any subsidy and zero-inflated count models produce the same basic nationality disadvantage (Appendix Tables A8–A9). Matching analyses using Coarsened Exact Matching (Iacus, King, and Porro, 2012) and Generalized Full Matching (Sävje, Higgins, and Sekhon, 2020) likewise show a negative nationality relationship among firms of similar size located in the same state, industry, and year (Appendix Table A10). These analyses cannot eliminate all concerns about unobserved differences between foreign and U.S. firms, but they indicate that the nationality gap is not driven solely by the most obvious compositional differences in the sample.

Average adjusted predictions from PPML models provide a more intuitive sense of the substantive difference on the original outcome scale. Holding the included covariates and fixed effects constant, U.S. firms are predicted to receive 0.471 subsidies per firm-year, compared with 0.194 for foreign firms, a difference of 0.277 awards, or 58.8%. Predicted subsidy amounts exhibit an even larger disparity: U.S. firms are predicted to receive an average of \$1.766 million per firm-year, compared with \$0.396 million for foreign firms, a difference of approximately \$1.371 million, or 77.6% (Appendix Table A4). Because PPML estimates

conditional means on the original scale and is more sensitive to large subsidy awards, these magnitudes differ from those obtained from OLS models using $\log(1 + y)$ outcomes. Both approaches nevertheless produce the same substantive conclusion. Taken together, these results support Hypothesis 1: foreign nationality constitutes a meaningful disadvantage in the allocation of U.S. industrial subsidies.

Foreign supplier dependence

Hypothesis 2 extends the argument to a second dimension of firm foreignness. If governments seek to retain the economic benefits of industrial subsidies within the domestic economy, firms that depend more heavily on foreign suppliers should receive less support. At the same time, the theory anticipates that supplier nationality should be less politically consequential than nationality or direct employment because upstream relationships are less visible and less readily attributable.

The results do not support the directional prediction in Hypothesis 2. As Table 2 shows, the coefficient on lagged foreign supplier share is positive rather than negative. Firms with greater foreign supplier exposure receive modestly more subsidy awards ($\beta = 0.097, p < 0.05$) and substantially larger subsidy amounts ($\beta = 0.733, p < 0.05$). Moving from an entirely domestic to an entirely foreign observed supplier base is associated with roughly a 10% increase in subsidy counts and approximately a doubling of subsidy amounts, although such a full-range change is uncommon in the data. The estimates remain nearly unchanged when foreign-firm status is included in the joint specification.

Hypothesis 2 is therefore not supported. The observational evidence provides no indication that governments systematically penalize firms for relying on foreign suppliers; if anything, the conditional association runs in the opposite direction. This result stands in sharp contrast to the robust disadvantage associated with foreign nationality. Although both headquarters location and supply chains connect firms to foreign economies, governments appear to distinguish sharply between these dimensions of firm foreignness.

The positive supplier coefficient itself should be interpreted cautiously. Appendix Table A7 shows that the relationship is sensitive to the geographic and industrial composition of firms. The unconditional association between foreign supplier exposure and subsidy allocation is negative, but it changes as state, year, and industry fixed effects are introduced. Supplier information is also available only for a comparatively small subset of firm-years. These patterns make it difficult to interpret the positive conditional coefficient as evidence that foreign sourcing itself causes firms to receive more government support. The stronger

conclusion is that the data provide little evidence of a systematic political penalty for foreign sourcing comparable to the penalty associated with foreign nationality.

This contrast is nevertheless informative for the broader theoretical argument. Nationality is highly visible and clearly identifies where control of the firm resides, whereas supplier relationships are more diffuse and less readily attributable to individual subsidy decisions. The results therefore suggest that the political boundary between domestic and foreign firms is drawn much more sharply around nationality than around the geography of upstream production. This interpretation does not explain the positive supplier coefficient itself, which was not predicted by the theory, but it helps account for the substantial asymmetry between the nationality and supply-chain results.

The results thus reveal a sharp difference between two dimensions of firm foreignness: foreign nationality carries a substantial penalty, whereas foreign supplier dependence does not. I next examine whether these relationships vary with labor intensity, testing whether employment-intensive economic activity changes how governments respond to foreign nationality and foreign sourcing.

When do foreign firms overcome the nationality penalty?

The preceding results establish a clear disadvantage associated with foreign nationality, but they also raise the central question motivating this paper: if foreign nationality is politically costly, why do governments nevertheless subsidize many foreign firms? Hypothesis 3 argues that the answer depends on firms' employment orientation. If labor-intensive production provides a more visible basis for emphasizing employment benefits, the nationality penalty should diminish as firms become more employment intensive.

Table 3 provides strong evidence consistent with this expectation. Labor intensity is positively associated with subsidy receipt, but the more important result is its interaction with foreign-firm status. The interaction between foreign status and logged labor intensity is positive and statistically significant ($\beta = 0.638$, $p < 0.01$), indicating that the negative association between foreign nationality and subsidy allocation becomes progressively smaller as labor intensity increases.

Because the interaction coefficient alone does not reveal the substantive size of this change, Figure 3 plots predicted subsidy counts for U.S. and foreign firms across the observed distribution of labor intensity. Among relatively capital-intensive firms, the nationality divide is pronounced: foreign firms receive substantially fewer subsidies than otherwise comparable U.S. firms. As labor intensity increases, however, this gap narrows considerably.

Table 3: Labor intensity and the effects of firm foreignness

	1	2	3	4
	Log Count of Industrial Subsidies			
Foreign firm	-0.116** (0.036)	-0.119** (0.036)		
Labor intensity (log)	0.430** (0.051)	0.247** (0.060)		
Foreign firm $\times$ Labor intensity (log)		0.638** (0.109)		
Foreign supplier			0.098** (0.038)	0.095* (0.039)
Supplier labor intensity (log)			0.945** (0.289)	0.935** (0.291)
Foreign supplier $\times$ Supplier labor intensity (log)				0.870 (3.185)
Firm size	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes

Note: OLS estimates with robust standard errors in parentheses.

All models include firm size and state, year, and 4-digit NAICS industry fixed effects.

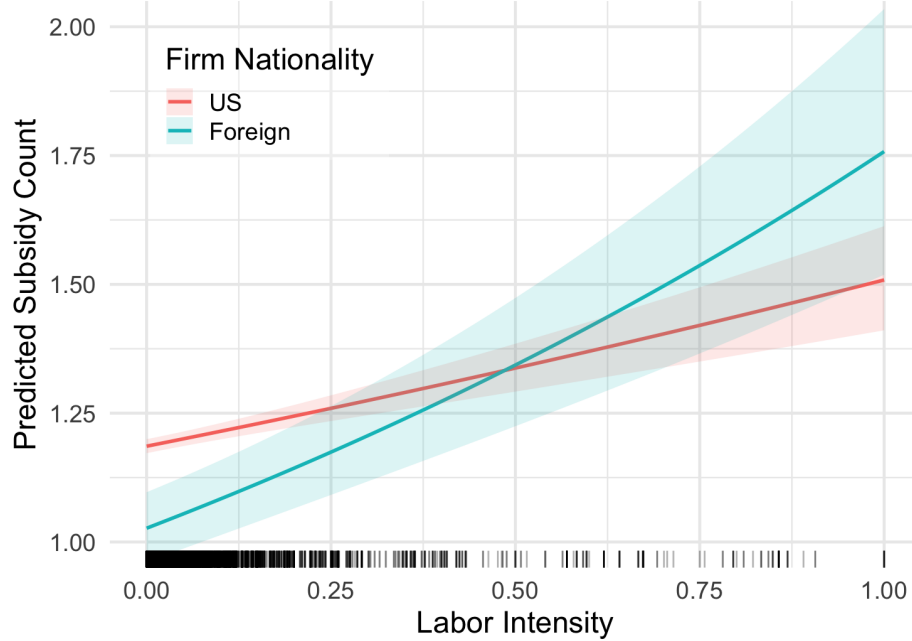
* $p < 0.05$, ** $p < 0.01$.

Around the middle of the observed distribution, the difference between foreign and U.S. firms becomes statistically indistinguishable from zero.

This pattern provides an answer to the apparent puzzle of foreign firms receiving subsidies under nationally oriented industrial policy. Governments do not appear to treat foreign nationality as an absolute disqualification. Instead, the nationality penalty is conditional on firms' employment orientation. Foreign firms whose production is relatively capital intensive retain a substantial disadvantage, while foreign firms for which employment constitutes a larger component of economic activity face a much smaller one. This pattern is consistent with employment providing one pathway through which foreign firms can overcome the political liability associated with nationality, although the data do not identify where firms' employees are located.

The interaction shows that employment orientation plays a different theoretical role from foreign nationality. It is not simply that labor-intensive firms receive more government support. If that were the entire story, labor intensity would have the same relationship with subsidy allocation regardless of nationality. Instead, its association is particularly strong

Figure 3: Predicted subsidy count by firm nationality and labor intensity



among foreign firms, precisely where policymakers face the greatest tension between supporting economic activity at home and directing public resources toward a foreign company. The interaction is therefore consistent with the argument that employment-oriented production makes the economic contribution of a foreign firm easier to identify and politically attribute.

Does the same logic extend beyond employment at the subsidized firm? Models 3 and 4 of Table 3 examine whether foreign supplier dependence becomes more consequential when the firms in those supplier networks are themselves labor intensive. Supplier labor intensity is positively associated with subsidy receipt, but its interaction with foreign supplier share is statistically indistinguishable from zero. There is therefore no evidence that governments respond differently to foreign sourcing depending on the employment intensity of the upstream supplier network.

The contrast between the two interactions is revealing. Employment intensity matters when it is attached directly to the firm receiving public support: greater labor intensity substantially attenuates the foreign-nationality penalty. The same pattern does not emerge for employment located indirectly in foreign supplier relationships. Together with the earlier supplier results, this suggests that governments place greater weight on employment at the subsidized firm than on the employment structure of its upstream production network.

These findings should nevertheless be interpreted as conditional associations rather than direct evidence of policymakers’ decision-making process. Labor intensity is not randomly assigned, and the measure captures the employment intensity of production rather than the exact number of U.S. jobs attributable to a particular subsidy. The estimates therefore do not imply that policymakers consciously trade a specific number of jobs against foreign nationality. They instead reveal a systematic allocation pattern consistent with Hypothesis 3: the nationality penalty is substantially weaker among firms for which employment represents a more prominent component of economic activity. The analysis also captures realized subsidy outcomes rather than applications alone, so differences may partly reflect firms’ decisions to seek support as well as governments’ allocation decisions.

Summary of Study 1

Study 1 reveals a differentiated rather than categorical response to firm foreignness. Foreign nationality is associated with fewer subsidies and lower total subsidy amounts, supporting Hypothesis 1. Yet this disadvantage is not uniform: it narrows considerably among firms with more employment-intensive production, consistent with Hypothesis 3. By contrast, governments do not appear to penalize reliance on foreign suppliers, and the supplier labor-intensity analysis provides no support for Hypothesis 4. Additional analyses also show that the nationality relationship varies across broad sectors: it is positive among tradable-goods firms but negative among service-sector firms (Appendix Table A13), indicating that the aggregate penalty should not be interpreted as uniform across the economy.

Taken together, the results support the central distinction developed in the theory. Firm nationality and foreign supplier reliance are two dimensions of foreignness, but only nationality carries a clear penalty. Employment orientation plays a different role: it is associated with a substantially weaker nationality disadvantage. Study 2 turns to the public to examine whether citizens make the same distinctions among ownership, workforce location, and supplier origin.

Study 2: Public Support for Industrial Subsidies

Research design

To test Hypotheses H1s–H4s, I field two preregistered survey experiments on nationally representative samples of U.S. adults recruited through YouGov, each targeting approximately

2,000 completed responses (4,000 respondents total). YouGov poststratification weights are applied throughout the confirmatory analyses. A pilot study conducted prior to the confirmatory fielding was used to refine the survey instrument; pilot responses are excluded from the analyses below.

Both experiments employ a 2×2 between-subjects design. Experiment 1 independently varies firm ownership and workforce composition. It tests whether an American workforce provides a domestic economic contribution that offsets the foreign-ownership penalty. Experiment 2 independently varies firm ownership and supplier origin. It tests whether citizens respond to ownership and supplier-based foreignness separately or treat them as reinforcing signals. Industry, annual revenue, and requested subsidy amount are held constant across conditions. Table 4 summarizes the experimental manipulations.

Table 4: Experimental conditions

Dimension	Domestic condition	Foreign condition
Ownership	American company headquartered in the United States	Foreign-owned company headquartered overseas
Workforce	About 2,000 American workers, approximately 80% of the total workforce	About 2,000 foreign workers outside the United States, approximately 80% of the total workforce
Supplier origin	Over 80% of parts and materials sourced from U.S.-based suppliers	Only 20% sourced from U.S.-based suppliers, with most parts and materials obtained from abroad

Note: Experiment 1 randomizes ownership and workforce composition; Experiment 2 randomizes ownership and supplier origin. Across all conditions, Nova Systems manufactures machines used to produce computer chips at U.S. factories, has approximately \$80 million in annual revenue, and applies for \$15 million in government support.

Respondents in the American-owned, American-workforce condition of Experiment 1 read the following vignette. The remaining conditions independently vary the ownership and workforce or supplier information summarized above. The complete wording of all eight experimental conditions appears in Appendix B.

Sample Vignette: The U.S. government is offering loans and grants to companies to strengthen American manufacturing. Nova Systems is an American company that makes machines used to produce computer chips at factories in the U.S. It is headquartered in the United States and employs about 2,000 American workers, which is roughly 80% of its total workforce. Nova is applying for \$15 million in government support and its annual revenue is about \$80 million.

Immediately after the vignette, respondents report their support for subsidizing the firm

on a five-point scale (1 = strongly oppose; 5 = strongly support), the paper’s primary outcome. They also evaluate how much subsidizing the firm would contribute to U.S. economic growth and employment and whether it would demonstrate that America comes before other nations. These secondary outcomes capture perceived economic contribution and national alignment and are examined later in the analysis.

Before the vignette, the survey also measures employment status, occupation, and nationalist orientation for the preregistered moderator analyses reported in Appendix B. Confirmatory models are estimated using survey-weighted OLS with YouGov poststratification weights and heteroskedasticity-robust (HC2) standard errors. The primary specifications include no additional covariates; demographic and political controls are added in secondary specifications as robustness checks.

Results

Main effects Pooling respondents across the two experiments, foreign ownership reduces subsidy support by 0.165 points on the five-point scale ($SE = 0.037$, $p < 0.001$). This pooled estimate supports Hypothesis 1s, although the experiment-specific results show that the ownership penalty is context-dependent.

Table 5: Pooled effect of foreign ownership on subsidy support

	Subsidy support
Foreign firm	−0.165*** (0.037)
Experiment 2	0.047 (0.037)

Note: Survey-weighted OLS estimates using the pooled sample. Experiment 1 is the reference design. HC2 robust standard errors are shown in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table 6 reports the experiment-specific main effects of ownership, workforce composition, and supplier origin on subsidy support. In Experiment 1, the foreign-ownership estimate is small and statistically indistinguishable from zero ($b = -0.070$, $p > 0.10$). In Experiment 2, by contrast, foreign ownership reduces support by approximately one-quarter of a point on the five-point scale ($b = -0.264$, $p < 0.001$). Thus, although the pooled analysis supports Hypothesis 1s, the experiment-specific estimates indicate that the ownership penalty is concentrated in the supplier-origin design.

The effects of workforce and supplier composition are considerably larger and more consistent. In Experiment 1, respondents are about half a point less supportive of subsidizing a firm that employs primarily foreign rather than American workers ($b = -0.511$, $p < 0.001$), supporting Hypothesis 2sa. In Experiment 2, sourcing primarily from foreign rather than U.S.-based suppliers reduces support by 0.445 points ($p < 0.001$), supporting Hypothesis 2sb. Both effects remain virtually unchanged after adding demographic and political controls.

Table 6: Effects of firm ownership, workforce, and supplier composition on subsidy support

	Experiment 1			Experiment 2		
	Weighted	+Controls	+Political	Weighted	+Controls	+Political
Foreign firm	-0.070 (0.052)	-0.065 (0.052)	-0.064 (0.052)	-0.264*** (0.052)	-0.274*** (0.051)	-0.282*** (0.051)
Foreign worker	-0.511*** (0.052)	-0.515*** (0.052)	-0.513*** (0.051)			
Foreign supplier				-0.445*** (0.052)	-0.442*** (0.051)	-0.436*** (0.051)

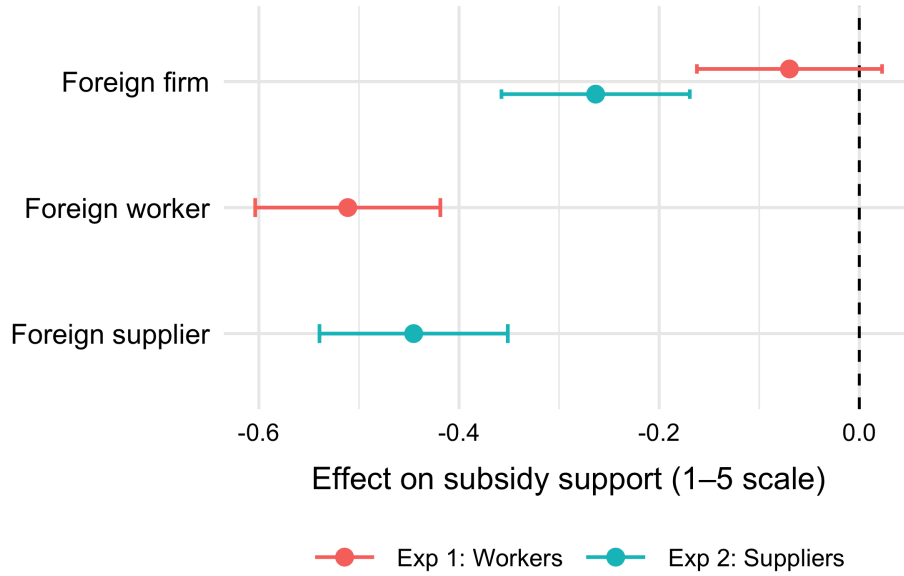
Note: Survey-weighted OLS estimates with YouGov poststratification weights. HC2 standard errors in parentheses. The outcome is subsidy support on a five-point scale. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Figure 4 makes the substantive contrast especially clear. The estimated penalties associated with a foreign workforce and foreign suppliers are substantially larger than the corresponding ownership effects. The workforce penalty is the largest, followed by supplier origin, while the ownership effect is smaller and varies across experiments. Citizens therefore distinguish between who owns a firm and where its economic activity takes place, placing substantial weight on the latter when deciding whether a firm deserves public support.

Do the firm characteristics condition one another? The interaction in Experiment 1 tests whether an American workforce compensates for foreign ownership. Consistent with Hypothesis 3s, the Foreign Firm $\times$ Foreign Worker coefficient is expected to be positive: the foreign-ownership penalty should be larger when the workforce is also foreign. The estimate is positive but small and statistically insignificant ($b = 0.088$, $p > 0.10$). There is therefore no evidence that the employment contribution becomes especially valuable when the firm is foreign-owned.

Experiment 2 tests a different relationship. Firm ownership and supplier origin represent two dimensions of foreignness, so Hypothesis 4s expects them to reinforce one another. The Foreign Firm $\times$ Foreign Supplier interaction is negative, as predicted, but small and statistically insignificant ($b = -0.064$, $p > 0.10$). Thus, respondents penalize foreign ownership and

Figure 4: Effects of ownership, workforce location, and supplier origin on subsidy support



Note: Points report survey-weighted OLS estimates from the primary specifications; horizontal bars indicate 95% confidence intervals. The outcome is support for subsidizing the firm on a five-point scale. Ownership is independently randomized in both experiments; workforce composition is randomized in Experiment 1 and supplier origin in Experiment 2.

foreign suppliers, but do not respond significantly more negatively when both characteristics appear together. The full interaction models are reported in Appendix Table B1.

Table 7 makes the substantive pattern clearer by decomposing these interactions. In Experiment 1, employing primarily foreign rather than American workers reduces support substantially among both American-owned firms ($b = -0.550$, $p < 0.001$) and foreign-owned firms ($b = -0.477$, $p < 0.001$), supporting the preregistered within-ownership hypotheses H4a and H4b reported in Appendix B. Experiment 2 shows the same pattern: sourcing primarily from foreign rather than American suppliers reduces support among both American-owned firms ($b = -0.401$, $p < 0.001$) and foreign-owned firms ($b = -0.456$, $p < 0.001$), supporting H7a and H7b. Workforce location and supplier origin therefore matter strongly, but neither effect differs sufficiently across ownership groups to support the predicted interactions.

Reading the comparison in the opposite direction leads to the same conclusion. In Experiment 1, the ownership penalty is statistically indistinguishable from zero whether the firm employs primarily American workers ($b = -0.114$) or foreign workers ($b = -0.019$). The American workforce therefore does not significantly attenuate the ownership penalty,

Table 7: Conditional Effects of Treatments on Subsidy Support

	Firm ownership	
	American-owned	Foreign-owned
Foreign workforce (vs. American)	-0.550***	-0.477***
Foreign suppliers (vs. American)	-0.401***	-0.456***
	Workforce composition	
	American workforce	Foreign workforce
Foreign ownership (vs. American)	-0.114	-0.019
	Supplier composition	
	American suppliers	Foreign suppliers
Foreign ownership (vs. American)	-0.248***	-0.325***

Note: Conditional treatment effects from survey-weighted linear models interacting firm ownership with workforce composition (Experiment 1) or supplier origin (Experiment 2). * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

contrary to Hypothesis 3s. In Experiment 2, foreign ownership reduces support under both sourcing conditions. The estimated penalty is somewhat larger when the firm relies primarily on foreign suppliers ($b = -0.325$) than when it relies primarily on American suppliers ($b = -0.248$), but the difference is not statistically significant. Hypothesis 4s is therefore also not supported.

The results therefore reveal an important distinction among the three firm characteristics. Citizens clearly reward an American workforce, but this employment contribution does not become especially valuable when a firm is foreign-owned. Citizens also penalize foreign suppliers, yet the two dimensions of foreignness do not significantly reinforce one another. Ownership, workforce location, and supplier origin largely shape subsidy preferences independently.

Two Evaluative Channels Why do citizens respond differently to ownership, workforce location, and supplier origin? Appendix Table 8 examines how the treatments shape perceived economic contribution and national alignment. Because these outcomes were measured after treatment, I interpret them as evidence about evaluative channels rather than as definitive causal mediation estimates. Foreign ownership reduces perceived national alignment in both experiments ($b = -0.148$, $p < 0.01$ in Experiment 1; $b = -0.244$, $p < 0.001$ in Experiment 2). Its relationship with perceived economic contribution is smaller and statistically insignificant in Experiment 1 ($b = -0.049$, $p > 0.10$), although it is negative and significant in Experiment 2 ($b = -0.178$, $p < 0.001$). Ownership therefore shapes national evaluations

more consistently than perceptions of economic contribution.

Table 8: Treatment Effects on Perceived Economic Contribution and National Alignment

	Experiment 1		Experiment 2	
	Economic contribution	National alignment	Economic contribution	National alignment
Foreign firm	-0.049 (0.047)	-0.148** (0.046)	-0.178*** (0.045)	-0.244*** (0.048)
Foreign worker	-0.260*** (0.047)	-0.423*** (0.046)		
Foreign supplier			-0.273*** (0.045)	-0.218*** (0.048)

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Survey-weighted OLS estimates with demographic and political covariates. HC2 robust standard errors are shown in parentheses. These post-treatment outcomes are interpreted as evaluative channels rather than definitive causal mediators.

Workforce and supplier composition operate more broadly. A foreign workforce reduces perceived economic contribution by 0.260 points and perceived national alignment by 0.423 points (both $p < 0.001$). Foreign sourcing similarly reduces perceived economic contribution by 0.273 points and national alignment by 0.218 points (both $p < 0.001$). This helps account for why workforce and supplier origin generate larger and more consistent effects on subsidy support than ownership alone. The heterogeneity analyses reinforce this interpretation: the ownership penalty becomes larger with nationalist orientation and conservative ideology in the supplier experiment, while the penalties associated with foreign workers and suppliers remain large across nearly all of the subgroups examined (Appendix Figures B1–B2).

Summary of Study 2

Study 2 shows that citizens evaluate potential subsidy recipients by considering where firms generate economic benefits, not simply who owns them. In the pooled sample, respondents are significantly less supportive of subsidizing a foreign-owned firm. The experiment-specific estimates indicate that this ownership penalty is concentrated in Experiment 2 and is not statistically distinguishable from zero in Experiment 1. Ownership nevertheless remains relevant across the two experiments: foreign ownership consistently lowers perceptions that a firm shares American interests and priorities, even when its effect on subsidy support varies across the experimental settings.

Workforce and supplier composition produce larger and more consistent effects. Respondents are substantially less supportive of firms that employ primarily foreign workers or rely primarily on foreign suppliers. These characteristics also reduce perceptions that the firm contributes to the U.S. economy and is aligned with American interests. The results therefore support Hypotheses 2sa and 2sb. Citizens recognize an American workforce as a direct domestic economic contribution and treat reliance on foreign suppliers as a separate form of foreign exposure.

The interaction and subgroup analyses provide no support for Hypotheses 3s and 4s. An American workforce does not systematically reduce the ownership penalty, and foreign ownership and foreign suppliers do not significantly reinforce one another. Instead, the effects of workforce and supplier location are broadly similar for American- and foreign-owned firms. Citizens appear to evaluate the three characteristics largely independently.

Taken together, the findings show that public support for industrial subsidies cannot be reduced to a simple preference for American-owned firms. Ownership provides one signal, particularly about national alignment. Workforce location indicates whether the focal firm makes a direct domestic employment contribution, while supplier origin captures a separate dimension of foreignness. Citizens respond to each characteristic, but do not consistently use one to revise their response to another.

Discussion

A Common Logic Beyond Nationality

Taken together, Studies 1 and 2 connect two sides of industrial policy: how governments allocate subsidies and how citizens evaluate potential recipients. Governments must not only select firms but also justify using public resources to support them. By examining both actual allocation and public support, the paper identifies a common political logic underlying the evaluation of globally organized firms.

The central finding across both studies is that nationality matters, but it is not sufficient. Governments and citizens distinguish firm nationality, employment, and supplier relationships rather than placing firms into a single domestic–foreign category. These characteristics also play different roles. Firm nationality and foreign supplier reliance represent two dimensions of foreignness. Employment represents a domestic economic contribution that can provide a reason to support a foreign firm. Both studies therefore show that the political evaluation of firms extends beyond where they are headquartered.

Study 1 demonstrates this logic in actual government allocation. Foreign firms receive fewer subsidies and lower total subsidy amounts than comparable U.S. firms, indicating that nationality creates a clear disadvantage. This disadvantage, however, becomes substantially smaller among more labor-intensive firms. The pattern is consistent with employment-oriented production providing policymakers with a visible and defensible reason to support a foreign firm. Foreign supplier reliance, by contrast, does not reduce government support. Governments therefore distinguish foreign corporate nationality from participation in foreign supply chains.

Study 2 finds a similar multidimensional logic among citizens. Respondents evaluate firm ownership separately from workforce location and supplier origin. They are less supportive of firms with foreign workforces or foreign suppliers, while the ownership effect is smaller and less consistent. Public support therefore does not depend only on whether a firm is American- or foreign-owned. Citizens also consider where the firm employs workers and sources inputs when deciding whether it deserves government support.

How Governments and Citizens Weigh Firm Characteristics

The two studies show broad agreement about which firm characteristics matter, although governments and citizens do not give each characteristic the same weight. Employment is important in both studies. In government allocation, labor intensity attenuates the disadvantage associated with foreign nationality. In the survey, an American workforce produces the largest increase in public support. These results identify employment as an especially important economic contribution across both government behavior and public opinion.

The combination of nationality and employment nevertheless differs somewhat across the studies. Government allocation is consistent with employment orientation mitigating the nationality penalty. Citizens strongly prefer an American workforce, but an American workforce does not significantly reduce their response to foreign ownership. Citizens appear to value domestic employment regardless of who owns the firm. This difference qualifies, rather than overturns, the common finding: employment matters to both governments and citizens, but they do not necessarily combine it with nationality in the same way.

Supplier relationships show a similar pattern. Governments do not systematically penalize firms for relying on foreign suppliers, while citizens respond negatively to foreign sourcing when it is made explicit. Yet both studies distinguish supplier relationships from firm nationality. Foreign sourcing is not treated as equivalent to foreign corporate control. The two groups differ in how much weight they place on supplier origin, not in whether it represents

a separate characteristic.

These differences may reflect the positions from which policymakers and citizens evaluate firms. Policymakers negotiate with firms and consider employment commitments, investment plans, supply-chain vulnerabilities, and strategic objectives. They also have incentives to emphasize outcomes for which they can claim political credit, especially job creation. Citizens generally possess less detailed information and may respond separately to visible information about ownership, workers, and suppliers. The experimental presentation of supplier origin may also make foreign sourcing more salient than it is in actual subsidy decisions.

The measures used in the two studies also differ. Study 1 measures labor intensity but cannot identify the location of every employee within a multinational firm. Its results concern employment orientation rather than directly measured domestic job creation. Experiment 1 directly varies whether the workforce is American or foreign. Study 1 measures supplier relationships through disclosed firm links, while Experiment 2 provides respondents with explicit information about supplier origin. The studies are therefore complementary rather than identical tests of the same relationship.

Industrial Policy Under Globalization

Combining the two studies is a central contribution of the paper. The observational analysis shows how governments actually distribute subsidies, while the experiments show how citizens evaluate comparable firm characteristics. The allocation data alone cannot establish whether the considerations reflected in government behavior are meaningful to the public. The experiments alone cannot show whether those considerations shape actual subsidy allocation. Together, they show that governments and citizens share a broad logic that looks beyond nationality, even though they sometimes weigh the relevant characteristics differently.

More broadly, industrial policy does not simply retreat from globalization. It attempts to direct globally organized firms toward national goals. Foreign firms can receive public support when their activities can be connected to employment and other visible economic benefits at home. Citizens also recognize these contributions rather than evaluating firms solely by nationality. Industrial policy therefore manages the tension between national objectives and global production by distinguishing among the different ways firms are connected to domestic and foreign economies.

Conclusion

This paper starts with a simple problem. Industrial policy serves national goals, but many of the firms that receive support operate across national borders. A foreign headquarters does not tell us where a firm employs workers, produces goods, or buys inputs. The results show that governments and citizens consider these activities when they evaluate firms. Foreign nationality still matters, but it does not determine whether a firm is seen as contributing to the domestic economy.

This finding contributes to firm-level research in international political economy. Existing studies often examine how firms with different interests influence trade and investment policy. This paper asks a different question: how do governments and citizens choose among firms? Firms differ not only in size, productivity, or sector. They also differ in where they create jobs and organize production. These differences affect whether a firm is considered an appropriate recipient of national economic policy. Firm-level IPE should therefore examine both how firms shape policy and how governments classify and treat different firms.

The paper also contributes to research on foreign direct investment and the liability of foreignness. This literature shows that foreign firms may face disadvantages because they have weaker political ties, less knowledge of local institutions, or less legitimacy in the host country (Zaheer, 1995; Johns and Wellhausen, 2021; Kim and Siegel, 2024). The findings here show that this liability is not fixed. Foreign firms remain disadvantaged on average, but the government-allocation results indicate that the disadvantage narrows as employment becomes a more prominent part of their production. Employment does not remove foreign nationality. It can, however, provide a domestic economic contribution that reduces some of its political costs.

This argument also helps explain the current backlash against globalization. The return of industrial policy does not mean that governments have stopped relying on global firms and supply chains. Instead, governments are trying to direct more of their benefits toward the domestic economy. They use subsidies, local-content rules, procurement policies, tariffs, and investment restrictions to influence where firms produce and employ workers. The result is not the end of globalization. It is a more selective form of globalization in which access to national markets and public support depends on what firms contribute at home.

One part of my future research will examine why governments choose different policies to pursue these goals. Tariffs protect domestic producers by making imports more expensive. Subsidies provide direct support to selected firms or investments. These policies may serve similar goals, but they distribute benefits and costs differently. Future work will examine

when governments choose subsidies rather than tariffs, which firms support each policy, and how firms' positions in global supply chains affect their preferences. This project will connect firm-level political preferences to the government's choice among industrial and trade policy tools.

Another part of my research will study how firms build political ties within the host country. Domestic subsidiaries, lobbying, and revolving-door connections may help foreign firms gain access to policymakers and present themselves as important local employers or investors. I also plan to use H-1B data to measure firms' reliance on foreign workers more directly. This will help distinguish labor intensity from the location and nationality of a firm's workforce. These projects will show how economic activity and political connections jointly shape access to government support.

My industrial policy research also extends to sectors in which technology and national security play a central role. Governments increasingly use subsidies, procurement, investment screening, and technology restrictions to support strategic industries and reduce dependence on geopolitical competitors. In these sectors, they may care about who controls a technology, whether a supplier comes from an ally, and whether a product works with existing national and allied systems. The question of which foreign firms serve national goals therefore becomes closely connected to security relationships and technical standards.

My research on arms trade develops this part of the industrial policy agenda. It examines whether NATO standards, interoperability requirements, and alliance relationships affect which weapons governments import and which foreign producers gain access to national defense markets. A product made by a foreign firm may become more acceptable when it is compatible with allied systems or produced in a trusted country. This project applies the paper's main question to technology and security: which connections make a foreign firm or product suitable for national support and procurement?

The broader implication is that headquarters nationality alone does not determine how a firm will be treated. Governments increasingly work with firms whose owners, workers, suppliers, and technologies are spread across countries. They must decide which forms of foreign exposure create political costs and which economic contributions justify public support. The same problem appears in industrial subsidies, foreign investment, trade policy, technology policy, and defense procurement. Understanding the global economy therefore requires studying how governments distinguish among these characteristics rather than combining them into a single domestic-foreign classification.

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Appendix A: Study 1 (Government Allocation) Supplementary Material

A.1 Descriptive statistics

Table A1: Summary statistics

	Count	Mean	SD	Min	Max
Annual subsidy count	151338	0.38	2.62	0.00	196.00
Annual subsidy amount	151338	1.16×10^6	4.00×10^7	0.00	8.83×10^9
U.S. firm	151338	0.77	0.42	0.00	1.00
Foreign firm	151338	0.23	0.42	0.00	1.00
Foreign supplier share	20996	0.04	0.18	0.00	1.00
Firm labor intensity	151338	0.01	0.02	0.00	1.00
Supplier labor intensity	20996	0.01	0.05	0.00	1.00
Annual revenue	151338	3.71×10^3	1.68×10^4	0.00	6.48×10^5
Number of employees	151338	10.03	44.06	0.00	2300.00

Note: The dependent variables and both labor intensity measures are log-transformed in the regression analyses. Annual revenue and employee count are standardized and combined into a composite firm-size control.

Table A2: Balance table: subsidized vs. non-subsidized firms

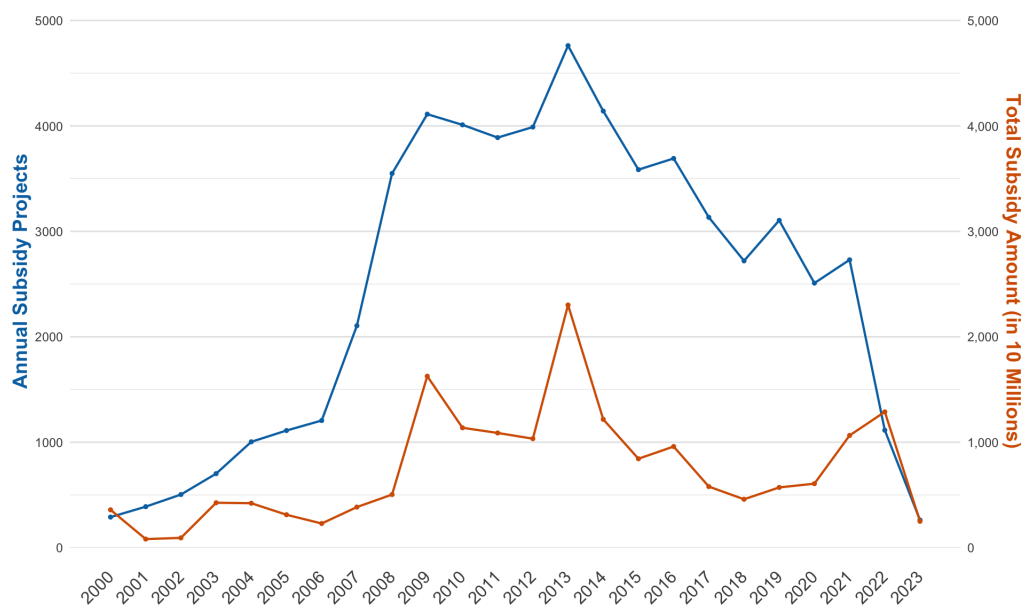
	Non-subsidized Firms	Subsidized Firms
U.S. firm	0.76	0.846
Foreign firm	0.238	0.154
Foreign supplier	0.039	0.032
Labor intensity	0.011	0.004
Supplier labor intensity	0.033	0.004
Annual revenue	2125.09	19222.45
Number of employees	6.208	47.442

Note: Subsidized firms are larger in size, with higher revenue and employee counts, than non-subsidized firms.

Table A3: Industrial subsidy awards by industry (NAICS 2-digit)

Code	Industry	Awards	Amount (\$B)
33	Manufacturing (Metal, Machinery, Electronics)	15,765	63.74
32	Manufacturing (Wood, Paper, Petroleum, Chemicals)	7,871	21.02
52	Finance and Insurance	5,053	8.36
51	Information	4,613	13.10
31	Manufacturing (Food, Textiles, Apparel)	3,984	4.43
45	Retail Trade	2,605	5.19
22	Utilities	1,975	25.24
48	Transportation and Warehousing	1,971	4.95
44	Retail Trade	1,894	0.97
99	Nonclassifiable	1,838	0.38
42	Wholesale Trade	1,408	0.55
21	Mining, Quarrying, and Oil & Gas Extraction	1,223	4.34
49	Transportation and Warehousing	1,095	0.53
54	Professional, Scientific, and Technical Services	903	1.66
53	Real Estate and Rental and Leasing	831	2.18
72	Accommodation and Food Services	800	0.66
62	Health Care and Social Assistance	604	0.61
56	Administrative and Support Services	449	0.13
71	Arts, Entertainment, and Recreation	242	0.36
23	Construction	160	0.08
61	Educational Services	97	0.11
81	Other Services	47	0.01
11	Agriculture, Forestry, Fishing, and Hunting	31	0.04
92	Public Administration	0	0.00

Figure A1: Annual count and amount of industrial subsidy projects



A.2 Distribution of dependent variables

Figure A2: Distribution of subsidy count (recipients only)

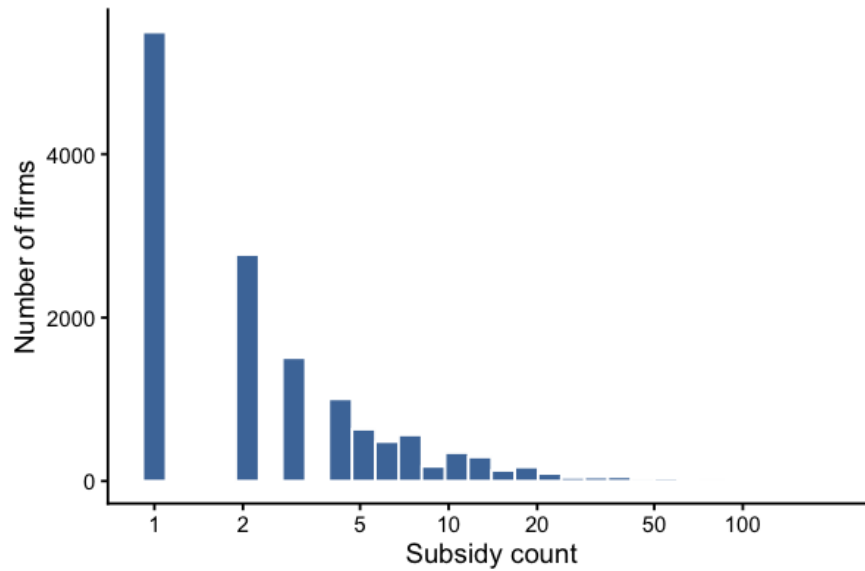
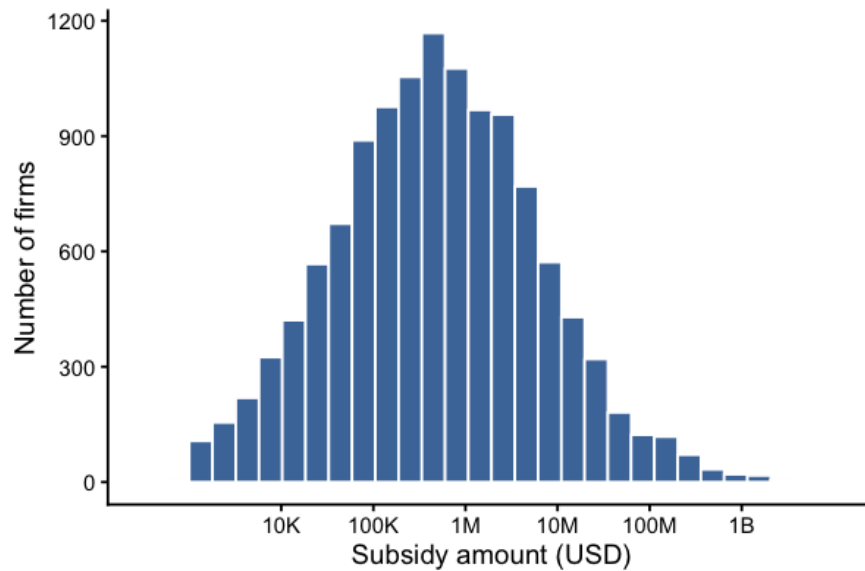


Figure A3: Distribution of subsidy amount (log scale, recipients only)



Note: Histograms are restricted to firms receiving at least one subsidy; x-axes are on a log scale.

A.3 Distribution of subsidies by firm nationality

Figure A4: Number of firms receiving industrial subsidies, by headquarters country

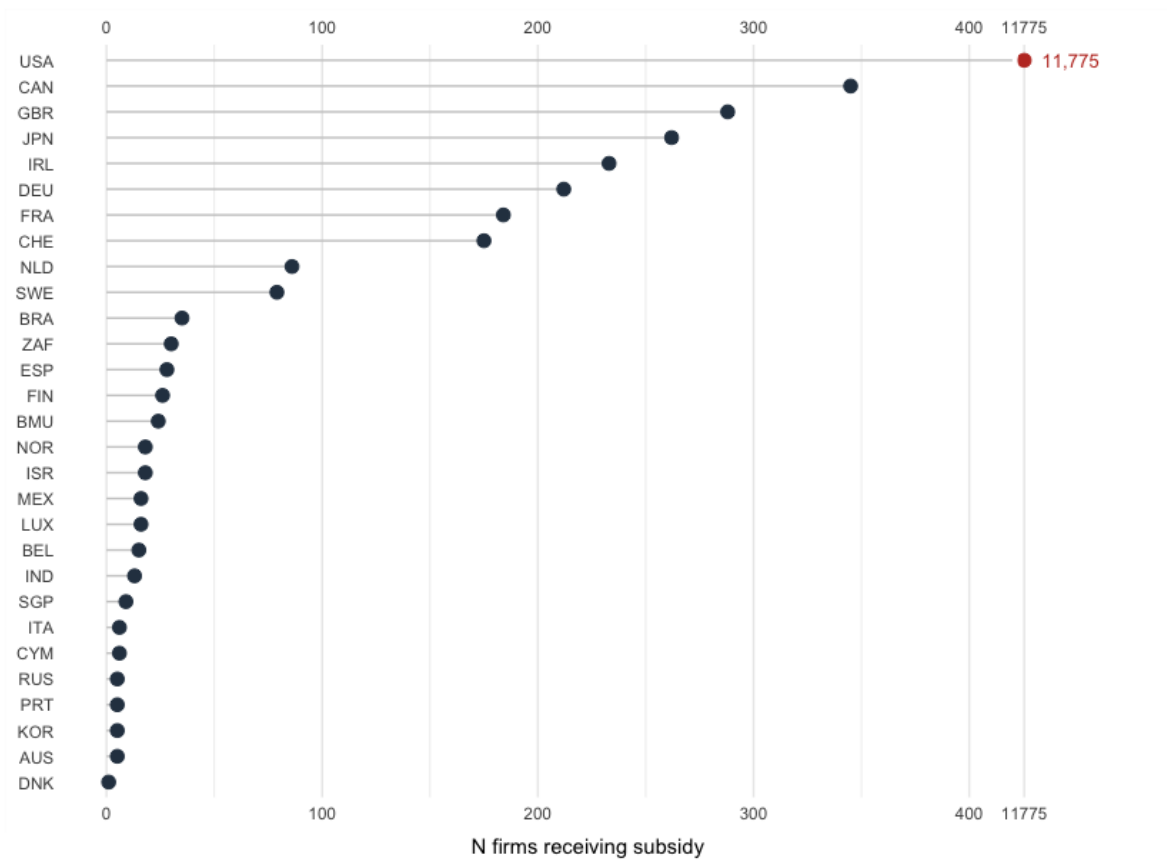
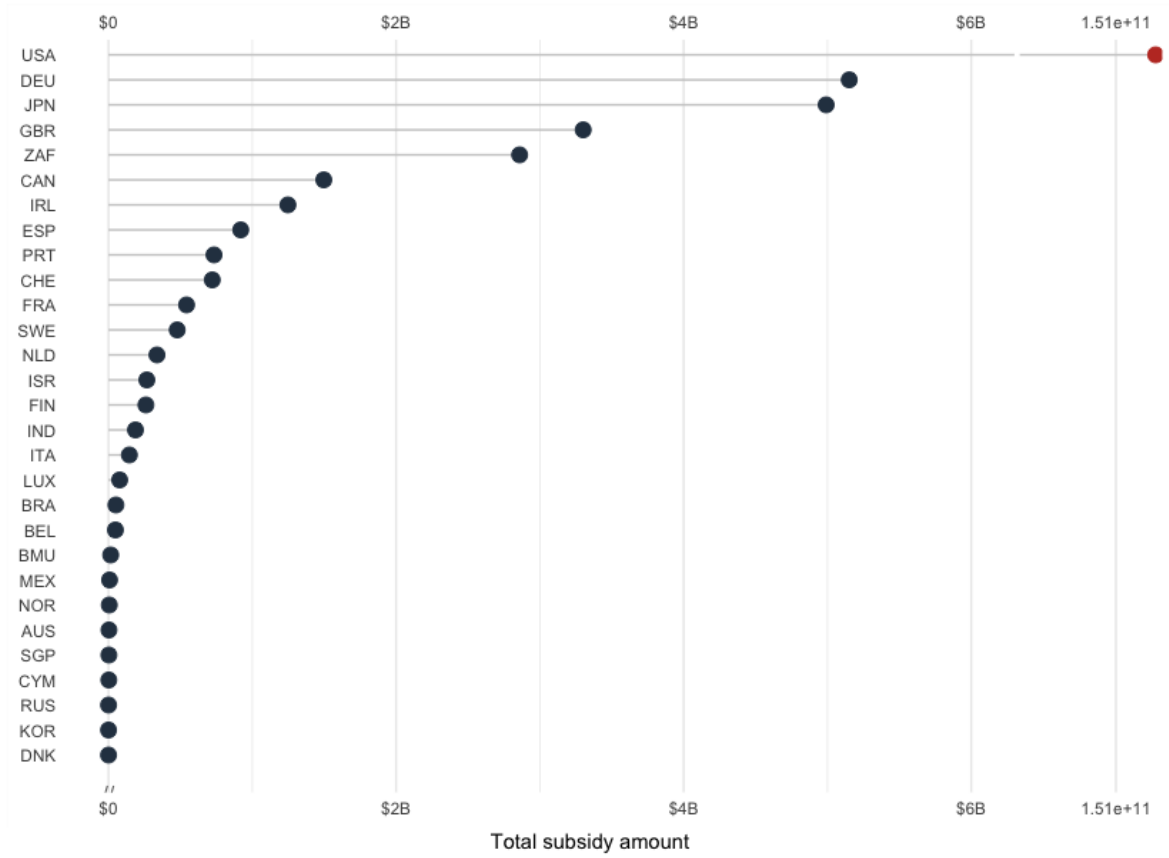
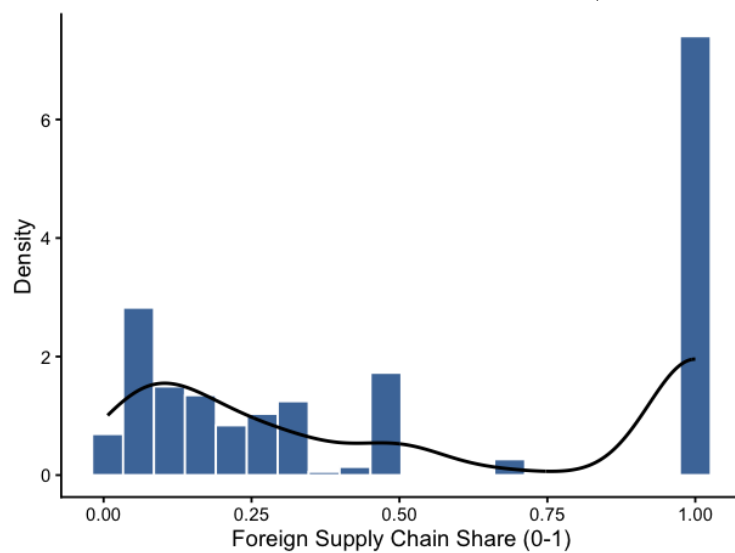


Figure A5: Total industrial subsidy amount, by headquarters country



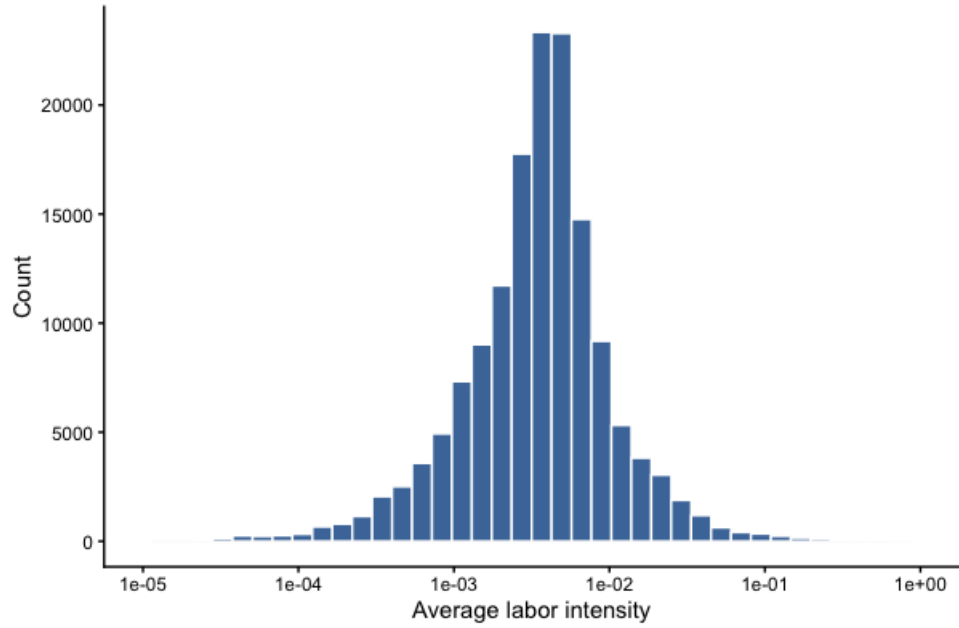
A.4 Distribution of foreign supplier and labor-intensity measures

Figure A6: Distribution of foreign supply chain share (non-zero observations)



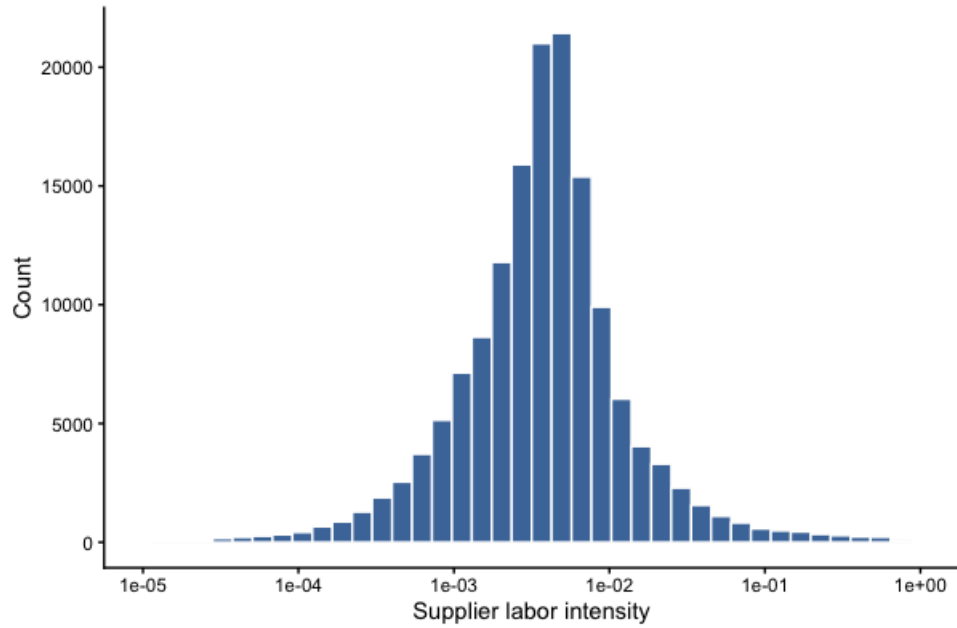
Note: Displays the distribution of Foreign supplier share among firms with positive foreign supplier shares only (8% of firm-years); the remaining 92% report zero and are omitted for visual clarity.

Figure A7: Distribution of average labor intensity



Note: Labor intensity is the ratio of employees to annual sales; both x-axes are log scale given right-skew.

Figure A8: Distribution of supplier labor intensity



A.5 Predicted outcomes

Table A4: Predicted subsidy outcomes by firm nationality

Outcome	U.S. firms	Foreign firms	Difference	% difference
Subsidy count	0.471	0.194	-0.277	-58.8%
Subsidy amount (\$ millions)	1.766	0.396	-1.371	-77.6%

Note: Average adjusted predictions from Poisson pseudo-maximum-likelihood models. For each firm-year, predictions are calculated under counterfactual U.S.-firm and foreign-firm status while retaining the observation's actual firm size, year, state, and four-digit NAICS industry. Predictions are then averaged across the estimation sample. Subsidy amounts are reported in millions of U.S. dollars. Percentage differences are calculated as $(\text{Foreign} - \text{U.S.})/\text{U.S.} \times 100$.

A.6 Robustness checks

Table A5: Robustness check: firm-clustered standard errors

	1	2	3
	Log Count of Industrial Subsidies		
Foreign firm	-0.115 (0.076)		-0.785*** (0.124)
Foreign supplier share _{t-1}		0.097 (0.085)	0.097 (0.085)
	Log Amount of Industrial Subsidies		
Foreign firm	-0.955** (0.310)		-6.015*** (1.332)
Foreign supplier share _{t-1}		0.733* (0.307)	0.739* (0.307)
Firm size	Yes	Yes	Yes
State FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes

Note: Standard errors clustered by firm (**gvkey**).

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Firm-clustered errors leave the nationality penalty negative and highly significant in the fully specified models for both subsidy count and amount; the count specification without the supplier control loses significance under clustering, though the coefficient magnitude is materially unchanged. The foreign supplier share coefficient is directionally consistent throughout, reaching significance in the amount models but not the count models.

Table A6: Robustness check: foreign nationality across FE specifications

	Count			Amount		
Foreign firm	-0.105** (0.002)	-0.122** (0.036)	-0.115** (0.036)	-0.895** (0.020)	-0.989** (0.305)	-0.955** (0.310)
Firm size	Yes	Yes	Yes	Yes	Yes	Yes
State FE	No	Yes	Yes	No	Yes	Yes
Year FE	No	Yes	Yes	No	Yes	Yes
NAICS 4-digit FE	No	No	Yes	No	No	Yes

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table A7: Robustness check: foreign suppliers across FE specifications

	Count			Amount		
Foreign suppliers (t-1)	-0.275** (0.040)	-0.006 (0.039)	0.097* (0.038)	-1.892** (0.315)	-0.104 (0.308)	0.733* (0.307)
Firm size	Yes	Yes	Yes	Yes	Yes	Yes
State FE	No	Yes	Yes	No	Yes	Yes
Year FE	No	Yes	Yes	No	Yes	Yes
NAICS 4-digit FE	No	No	Yes	No	No	Yes

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

The foreign-nationality penalty is robust across progressively saturated fixed-effects specifications. Foreign supplier exposure is more sensitive to specification: its unconditional negative association disappears with state and year fixed effects and reverses to significantly positive once industry fixed effects are added. The stronger conclusion is therefore that the data provide little evidence of a systematic supplier penalty comparable to the nationality penalty.

The linear probability model (Table A8) and zero-inflated negative binomial model (Table A9) both confirm the baseline pattern—a significant foreign-nationality penalty and a null-to-positive foreign-supplier association—indicating the results are not an artifact of the log-linear functional form or of how zero-subsidy observations are treated.

Table A8: Robustness check: binary subsidy-receipt outcome

	Subsidy receipt (binary)		
Foreign firm	-0.070** (0.024)		-0.368** (0.097)
Foreign suppliers (t-1)		0.057* (0.022)	0.057* (0.022)
Firm size	Yes	Yes	Yes
State FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
NAICS 4-digit FE	Yes	Yes	Yes

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table A9: Robustness check: zero-inflated negative binomial models

	Subsidy Count		
	(1)	(2)	(3)
Foreign firm	-1.022+ (0.573)		-2.602** (0.863)
Foreign suppliers (t-1)		0.156 (0.169)	0.161 (0.169)
Firm size	Yes	Yes	Yes
State FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
NAICS 4-digit FE	Yes	Yes	Yes

Note: Entries are coefficients from the count equation.

+ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

A.7 Matching analysis

Table A10: Coarsened exact matching and generalized full matching

	CEM	GFM
Foreign firm	-0.074*** (0.005)	-0.126*** (0.026)
Firm size	Yes	Yes
State FE	Yes	Yes
Year FE	Yes	Yes
Industry FE	Yes	Yes
SEs clustered at the firm level. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.		

Using Coarsened Exact Matching (Iacus, King, and Porro, 2012) and Generalized Full Matching (Sävje, Higgins, and Sekhon, 2020) to compare firms of similar size within the same state, industry, and year, the nationality penalty remains significant among observationally comparable firms, indicating that the baseline gap is not driven solely by observable firm characteristics. Following Ho et al. (2007), I treat matching as a preprocessing step rather than as a substitute for the outcome analysis.

A.8 Perceived foreignness and firm names

Firm nationality is often inferred through observable cues, such as company names, rather than legal headquarters alone. I classify firm names as English-sounding or foreign-sounding using OpenAI-based language detection combined with additional linguistic classification; approximately 13% of firms have foreign-sounding names, and more than one-quarter of firms with English-sounding names are not headquartered in the United States.

Table A11: Foreign-sounding firm names and subsidy allocation

	Full sample		U.S. firms only	
	1	2	3	4
Foreign-sounding firm	-0.016** (0.002)	-0.193** (0.022)	-0.004+ (0.002)	-0.075** (0.029)
Firm size	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes

+ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Firms with foreign-sounding names receive significantly less industrial support, even restricting to U.S.-headquartered firms, suggesting subsidy allocation is shaped by perceived as well as legal national identity.

A.9 Export and import exposure

Table A12: Industrial subsidies and industry export/import exposure

	Log Count of Subsidy		Log Amount of Subsidy	
	(1)	(2)	(3)	(4)
Export	0.050** (0.008)	0.015** (0.005)	0.311** (0.051)	0.107** (0.033)
Import	0.043** (0.009)	-0.017** (0.005)	0.180** (0.057)	-0.182** (0.037)
Industry size control	No	Yes	No	Yes
NAICS two-digit FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	5,979	5,979	5,979	5,979

Note: The unit of analysis is the NAICS four-digit industry-year.

Models include year and NAICS two-digit fixed effects; robust standard errors are in parentheses.

* $p < 0.05$, ** $p < 0.01$.

A.10 Tradable-goods and service-sector firms

Table A13: Industrial subsidies for tradable-goods and service-sector firms

	Tradable Goods				Services			
	Log Count		Log Amount		Log Count		Log Amount	
Foreign firm	0.119**		1.298**		-0.117**		-1.318**	
	(0.020)		(0.210)		(0.024)		(0.326)	
Foreign supplier share _{t-1}	-0.001		-0.128		0.033		1.099*	
	(0.030)		(0.386)		(0.037)		(0.510)	
Firm size	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

* $p < 0.05$, ** $p < 0.01$.

Note: Tradable goods include NAICS 11, 21, and 31–33. Services include NAICS 51–56, 61, and 62. All models include firm size and state, year, and industry fixed effects. Heteroskedasticity-robust standard errors are in parentheses.

A.11 Allocation by level of government

Table A14: Subsidy allocation by level of government in the United States

	All levels		Local		State		Federal	
U.S. firm	0.021*	0.085**	-0.031+	-0.024	0.038**	0.088**	0.072	0.310*
	(0.009)	(0.019)	(0.016)	(0.033)	(0.010)	(0.021)	(0.048)	(0.137)
U.S. supplier share _{t-1}	-0.066*			0.035		0.024		-0.791**
	(0.028)			(0.037)		(0.019)		(0.199)
Firm size	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	13,989	7,238	2,744	1,394	9,898	5,294	1,297	514

Note: Columns report $\log(1 + y)$ subsidy outcomes for awards from all, local, state, and federal programs.

All models include firm size and year and industry fixed effects. Heteroskedasticity-robust standard errors are in parentheses.

+ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$.

Appendix B: Study 2 (Public Support) Supplementary Material

B.1 Full vignette text

Respondents were randomly assigned to one of four conditions within each experiment. Experiment 1 independently varied firm ownership (American vs. foreign-owned) and workforce location (primarily American vs. primarily foreign). Experiment 2 independently varied firm ownership and supplier origin (primarily U.S.-based vs. primarily foreign). The full text of each experimental condition is reproduced below.

Experiment 1: Ownership $\times$ Workforce

American-owned firm $\times$ American workforce

The U.S. government is offering loans and grants to companies to strengthen American manufacturing.

Nova Systems is an American company that makes machines used to produce computer chips at factories in the U.S. It is headquartered in the United States and employs about 2,000 American workers, which is roughly 80% of its total workforce.

Nova is applying for \$15 million in government support and its annual revenue is about \$80 million.

American-owned firm $\times$ Foreign workforce

The U.S. government is offering loans and grants to companies to strengthen American manufacturing.

Nova Systems is an American company that makes machines used to produce computer chips at factories in the U.S. It is headquartered in the United States but employs about 2,000 foreign workers outside the U.S., which is roughly 80% of its total workforce.

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Foreign-owned firm $\times$ American workforce

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Nova is applying for \$15 million in government support and its annual revenue is about \$80 million.

Foreign-owned firm × Foreign workforce

The U.S. government is offering loans and grants to companies to strengthen American manufacturing.

Nova Systems is a foreign-owned company that makes machines used to produce computer chips at factories in the U.S. It is headquartered overseas and employs about 2,000 foreign workers outside the U.S., which is roughly 80% of its total workforce.

Nova is applying for \$15 million in government support and its annual revenue is about \$80 million.

Experiment 2: Ownership × Supplier Origin

American-owned firm × U.S.-based suppliers

The U.S. government is offering loans and grants to companies to strengthen American manufacturing.

Nova Systems is an American company that makes machines used to produce computer chips at factories in the U.S. It is headquartered in the United States and sources over 80% of its parts and materials from U.S.-based suppliers.

Nova is applying for \$15 million in government support and its annual revenue is about \$80 million.

American-owned firm × Foreign suppliers

The U.S. government is offering loans and grants to companies to strengthen American manufacturing.

Nova Systems is an American company that makes machines used to produce computer chips at factories in the U.S. It is headquartered in the United States but sources only 20% of its parts and materials from U.S.-based suppliers, obtaining most of its parts and materials from abroad.

Nova is applying for \$15 million in government support and its annual revenue is about \$80 million.

Foreign-owned firm × U.S.-based suppliers

The U.S. government is offering loans and grants to companies to strengthen American manufacturing.

Nova Systems is a foreign-owned company that makes machines used to produce computer chips at factories in the U.S. It is headquartered overseas but sources over 80% of its parts and materials from U.S.-based suppliers.

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Nova is applying for \$15 million in government support and its annual revenue is about \$80 million.

Outcome measures

Immediately after the vignette, respondents reported their support for providing the firm with a government subsidy. This measure constitutes the primary outcome in both experiments and is coded on a five-point scale from 1 (strongly oppose) to 5 (strongly support). Respondents were asked:

Would you support or oppose the U.S. government providing a subsidy to Nova Systems?

1. Strongly oppose
2. Somewhat oppose
3. Neither support nor oppose
4. Somewhat support
5. Strongly support

Respondents then evaluated the firm on two dimensions used to examine the mechanisms underlying subsidy preferences. The first, perceived economic contribution, captures the extent to which respondents expect subsidizing the firm to contribute to U.S. economic growth and employment. Respondents were asked:

How much would subsidizing Nova Systems contribute to economic growth and employment in the United States?

1. Not at all
2. A little
3. A moderate amount
4. A lot
5. A great deal

Responses are coded from 1 (not at all) to 5 (a great deal).

The second, perceived national alignment, captures whether respondents view subsidizing the firm as consistent with prioritizing American national interests. Respondents were asked:

Do you agree or disagree that subsidizing Nova Systems would demonstrate that America comes before other nations?

1. Strongly disagree
2. Somewhat disagree
3. Neither agree nor disagree
4. Somewhat agree
5. Strongly agree

Responses are coded from 1 (strongly disagree) to 5 (strongly agree).

Pre-randomization moderators

Before the experimental vignette, respondents answered three questions used in the moderator analyses. These measures capture respondents' occupational background and two distinct dimensions of nationalist orientation.

The first measure captures respondents' occupational background. Respondents were asked:

Which category best describes your current or most recent job?

1. Blue-collar (e.g., manufacturing, construction, transportation, maintenance)
2. White-collar (e.g., office work, management, professional services)
3. Other (please specify)
4. Never worked for pay

For the occupational moderator analysis, respondents are classified according to their reported occupational category.

The remaining two measures capture distinct dimensions of nationalist orientation. The first, national superiority, captures the belief that American characteristics or ways of life are preferable to those of other countries. Respondents were asked:

To what extent do you agree or disagree with the following statement:

“The world would be a better place if people from other countries were more like Americans.”

1. Strongly disagree
2. Somewhat disagree
3. Neither agree nor disagree
4. Somewhat agree
5. Strongly agree

Responses are coded from 1 (strongly disagree) to 5 (strongly agree), with higher values indicating stronger national superiority.

The second, national priority, captures the belief that the U.S. government should prioritize Americans over the interests of people in other countries. Respondents were asked:

To what extent do you agree or disagree with the following statement:

“The U.S. government should worry only about Americans when it makes decisions. Whether its decisions negatively impact people from other countries is not important.”

1. Strongly disagree
2. Somewhat disagree
3. Neither agree nor disagree
4. Somewhat agree
5. Strongly agree

Responses are coded from 1 (strongly disagree) to 5 (strongly agree), with higher values indicating stronger national priority. The two nationalism measures are analyzed separately rather than combined into a single index.

B.2 Full hypothesis set

The study preregistered the following confirmatory hypotheses. H1 concerns firm ownership and is evaluated across both experiments. H2–H4b concern workforce composition in Experiment 1, while H5–H7b concern supplier composition in Experiment 2.

The main text groups these preregistered expectations by concept. Main-text Hypotheses 2sa and 3s correspond to preregistered H2 and H3, respectively. Main-text Hypotheses 2sb and 4s correspond to preregistered H5 and H6.

Survey Hypothesis H1. The public will be less supportive of providing subsidies to foreign-owned firms than to American-owned firms.

Experiment 1: Workforce composition

Survey Hypothesis H2. The public will be less supportive of providing subsidies to firms that employ primarily foreign workers than to firms that employ primarily American workers.

Survey Hypothesis H3. The negative effect of foreign ownership on public support for subsidizing the firm will be smaller when the firm employs primarily American workers than when it employs primarily foreign workers.

Survey Hypothesis H4a. Among foreign-owned firms, employing primarily American workers will increase public support for subsidies relative to employing primarily foreign workers.

Survey Hypothesis H4b. Among American-owned firms, employing primarily foreign workers will reduce public support for subsidies relative to employing primarily American workers.

Experiment 2: Supplier composition

Survey Hypothesis H5. The public will be less supportive of providing subsidies to firms that source primarily from foreign suppliers than to firms that source primarily from American suppliers.

Survey Hypothesis H6. The negative effect of foreign ownership on public support for subsidizing the firm will be smaller when the firm sources primarily from American suppliers than when it sources primarily from foreign suppliers.

Survey Hypothesis H7a. Among foreign-owned firms, sourcing primarily from American suppliers will increase public support for subsidies relative to sourcing primarily from foreign suppliers.

Survey Hypothesis H7b. Among American-owned firms, sourcing primarily from foreign suppliers will reduce public support for subsidies relative to sourcing primarily from American suppliers.

Exploratory moderators and mediators

The preregistration also specified exploratory and explanatory expectations intended to distinguish between the material-benefit and national-identity mechanisms.

Material-benefit mechanism.

Moderator 1. The domestic-workforce premium will be larger among blue-collar than white-collar workers.

Moderator 2. The domestic-workforce premium will be smaller among retirees and economically inactive respondents than among active workers.

Mediator 1. The domestic-workforce effect will operate through perceived economic contribution.

National-identity mechanism.

Moderator 3. The domestic-ownership premium will increase with nationalist orientations, measured separately as national superiority and national priority.

Mediator 2. The domestic-ownership effect will operate through perceived national alignment.

B.3 Interaction analysis

Table B1: Interaction models: moderating effect of workforce/supplier composition

	Experiment 1			Experiment 2		
	Weighted	+Controls	+Political	Weighted	+Controls	+Political
Foreign firm	-0.113 (0.074)	-0.109 (0.074)	-0.105 (0.074)	-0.232** (0.072)	-0.244*** (0.072)	-0.247*** (0.072)
Foreign worker	-0.554*** (0.072)	-0.559*** (0.072)	-0.555*** (0.072)			
Foreign firm $\times$ Foreign worker	0.088 (0.103)	0.089 (0.103)	0.083 (0.103)			
Foreign supplier				-0.413*** (0.072)	-0.412*** (0.072)	-0.400*** (0.072)
Foreign firm $\times$ Foreign supplier				-0.064 (0.104)	-0.060 (0.103)	-0.071 (0.103)

+ p<0.1, * p<0.05, ** p<0.01, *** p<0.001. Robust (HC2) SEs in parentheses.

B.4 Heterogeneity by moderators

Blue-collar occupation significantly moderates the foreign-worker penalty in Experiment 1 ($b = 0.286$, $p < 0.01$), but in the direction opposite Moderator 1's prediction: blue-collar respondents penalize foreign-worker firms less, not more, than white-collar respondents; blue-collar status does not moderate either treatment in Experiment 2. Consistent with Moderator 3, nationalist orientation and conservative ideology both significantly widen the ownership penalty in the supplier design; neither reaches significance for the ownership effect in the workforce design, where the ownership main effect is itself close to zero, though both moderate the foreign-worker penalty in the expected direction. An exploratory partisan pattern is also worth noting: in the workforce design, the ownership penalty is concentrated among independents ($b = -0.174$, $p < 0.05$) and significantly smaller among Democrats (interaction $b = 0.312$, $p < 0.01$), with no comparable attenuation among Republicans.

Table B2: Moderators of foreign firm and foreign worker effects (Experiment 1)

	x Blue-collar	x Nationalism	x Ideology	x Partisanship
Foreign firm	-0.053 (0.054)	-0.055 (0.047)	-0.073 (0.050)	-0.174* (0.082)
Blue-collar	-0.125 (0.099)			
Foreign worker	-0.599*** (0.054)	-0.526*** (0.047)	-0.562*** (0.050)	-0.554*** (0.081)
Foreign firm $\times$ Blue-collar	0.011 (0.109)			
Blue-collar $\times$ Foreign worker	0.286** (0.109)			
Nationalism		0.044 (0.036)		
Foreign firm $\times$ Nationalism		-0.057 (0.040)		
Nationalism $\times$ Foreign worker		-0.052 (0.040)		
Ideology			0.081* (0.039)	
Foreign firm $\times$ Ideology			-0.075+ (0.044)	
Ideology $\times$ Foreign worker			-0.087* (0.044)	
Democrat				-0.112 (0.103)
Republican				0.100 (0.107)
Foreign firm $\times$ Democrat				0.312** (0.120)
Foreign firm $\times$ Republican				0.010 (0.124)
Democrat $\times$ Foreign worker				0.130 (0.120)
Republican $\times$ Foreign worker				-0.104 (0.124)

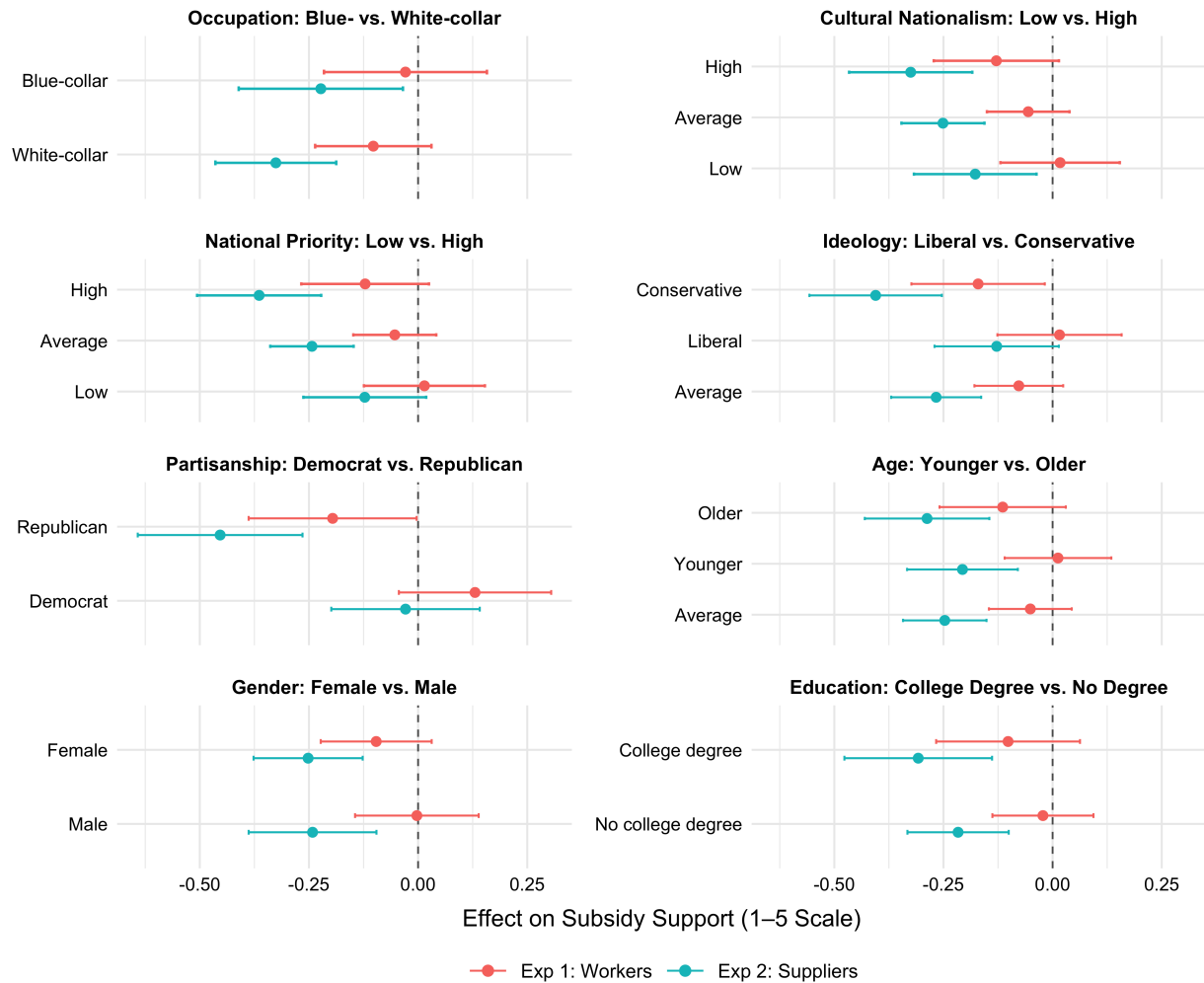
+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Robust (HC2) SEs in parentheses. Nationalism and ideology are mean-centered continuous moderators; reference category for partisanship is Independent.

Table B3: Moderators of foreign firm and foreign supplier effects (Experiment 2)

	x Blue-collar	x Nationalism	x Ideology	x Partisanship
Foreign firm	-0.261*** (0.055)	-0.249*** (0.048)	-0.274*** (0.051)	-0.268** (0.090)
Blue-collar	0.021 (0.098)			
Foreign supplier	-0.434*** (0.055)	-0.433*** (0.048)	-0.446*** (0.051)	-0.399*** (0.090)
Foreign firm $\times$ Blue-collar	0.032 (0.109)			
Blue-collar $\times$ Foreign supplier	0.010 (0.109)			
Nationalism		0.012 (0.036)		
Foreign firm $\times$ Nationalism		-0.082* (0.040)		
Nationalism $\times$ Foreign supplier		0.078+ (0.041)		
Ideology			0.116** (0.039)	
Foreign firm $\times$ Ideology			-0.111* (0.044)	
Ideology $\times$ Foreign supplier			-0.022 (0.044)	
Democrat				-0.012 (0.107)
Republican				0.336** (0.113)
Foreign firm $\times$ Democrat				0.212+ (0.123)
Foreign firm $\times$ Republican				-0.188 (0.130)
Democrat $\times$ Foreign supplier				-0.066 (0.124)
Republican $\times$ Foreign supplier				-0.070 (0.130)

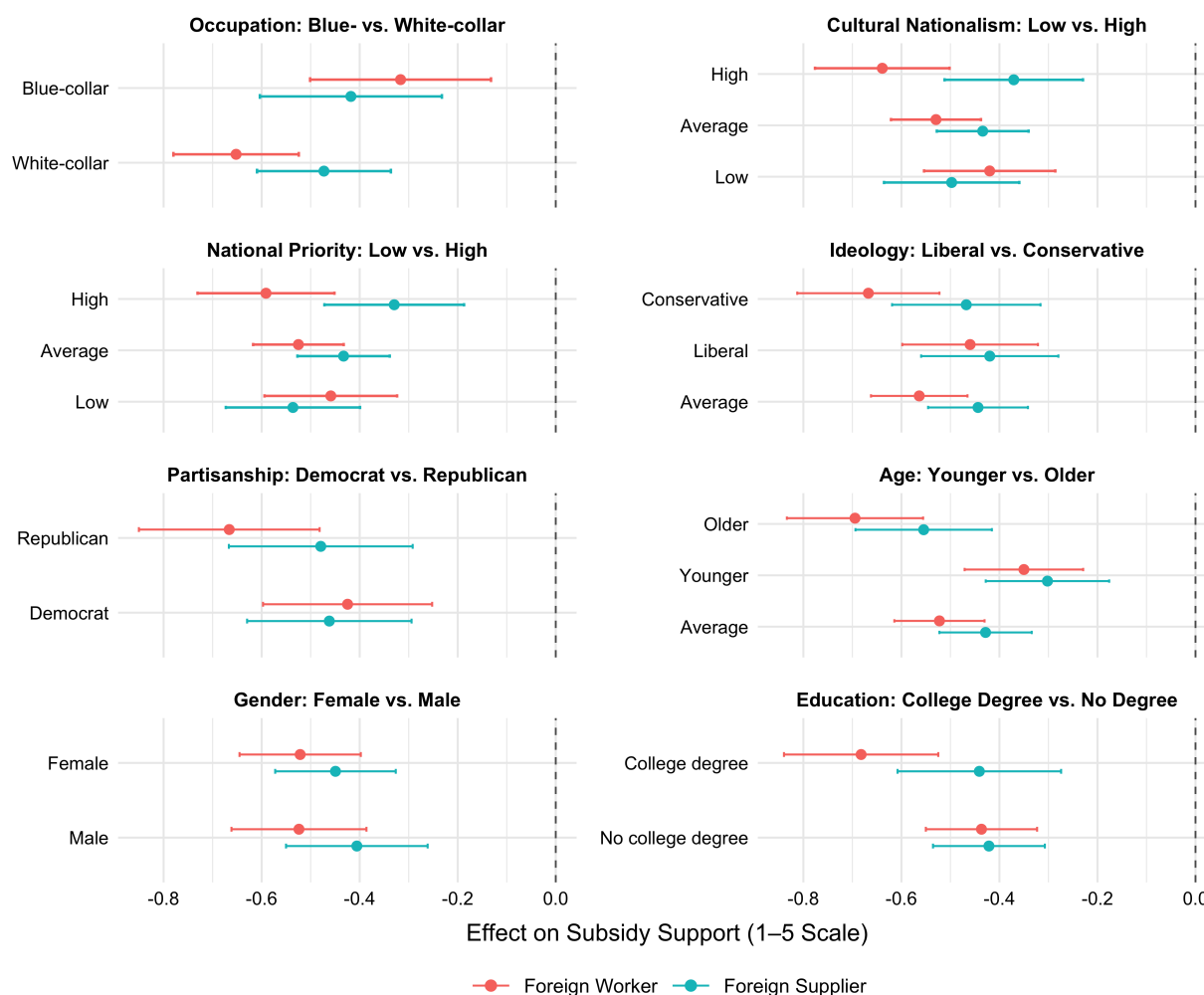
+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Robust (HC2) SEs in parentheses.

Figure B1: Heterogeneous treatment effects of foreign ownership



Note: Points show the estimated effect of foreign ownership on subsidy support across representative values of each moderator (binary moderators: 0/1; continuous moderators: mean $\pm$ 1 SD), with 95% CIs (HC2-robust SEs). Filled points/thick intervals are primary theoretical moderators; hollow points/thin intervals are exploratory controls.

Figure B2: Heterogeneous treatment effects of foreign worker and supplier ties



Note: Points show the estimated effect of foreign worker ties (Experiment 1) and foreign supplier ties (Experiment 2) on subsidy support across representative values of each moderator, with 95% CIs (HC2-robust SEs). Filled points/thick intervals are primary theoretical moderators; hollow points/thin intervals are exploratory controls.

Heterogeneity in treatment effects

Figures B1 and B2 examine whether the experimental treatments have different effects across respondent characteristics. Points report estimated treatment effects on the five-point subsidy-support scale at representative moderator values, with 95% confidence intervals. For binary moderators, estimates are reported for each category. For continuous moderators, estimates are reported at the mean and approximately one standard deviation above and below the mean. Because these analyses are exploratory and involve multiple comparisons, the results should be interpreted as evidence of broad patterns rather than as separate confirmatory tests.

Figure B1 shows that heterogeneity in the foreign ownership effect is concentrated primarily in Experiment 2. In Experiment 1, foreign ownership generally has a small and statistically indistinguishable effect across occupations, levels of nationalism, and demographic groups. There is some political differentiation: the ownership penalty is larger among more conservative respondents, while Democrats appear less opposed to foreign ownership than Independents. These differences, however, do not overturn the overall conclusion that ownership had little independent effect in the workforce experiment.

In Experiment 2, by contrast, foreign ownership reduces subsidy support across most respondent groups. The penalty becomes larger as nationalism and conservative ideology increase. It is also smaller among Democrats and larger among Republicans, relative to Independents, although not all subgroup differences are statistically precise. The ownership penalty is otherwise fairly consistent across occupation, age, gender, and education. Taken together, these results suggest that responses to foreign ownership are partly structured by national and partisan predispositions, especially when the alternative dimension of firm foreignness concerns supply chains.

Figure B2 reveals a different pattern. Foreign workers and foreign suppliers reduce subsidy support across every subgroup examined, and nearly all estimated effects remain statistically distinguishable from zero. Opposition to these forms of foreign economic activity is therefore not confined to nationalist, conservative, or Republican respondents. The most notable exception to this general uniformity is occupation: the foreign-workforce penalty is substantially smaller among blue-collar respondents than among other respondents. This result runs contrary to the expectation that blue-collar respondents would react more negatively to foreign employment. One possible interpretation is that these respondents place greater value on manufacturing activity itself, even when the firm's workforce is foreign. The foreign-workforce penalty also becomes larger among more nationalist and conserva-

tive respondents. Differences in the foreign-supplier effect are generally smaller and less systematic.

Overall, the heterogeneous-effects analyses reinforce the distinction among ownership, workforce location, and supplier origin. Opposition to foreign ownership varies more with nationalism, ideology, and partisanship, whereas the penalties for foreign workers and suppliers are broad-based across political and demographic groups. The latter pattern indicates that respondents' concern with where firms create jobs and source inputs extends beyond conventional ideological or partisan divisions.